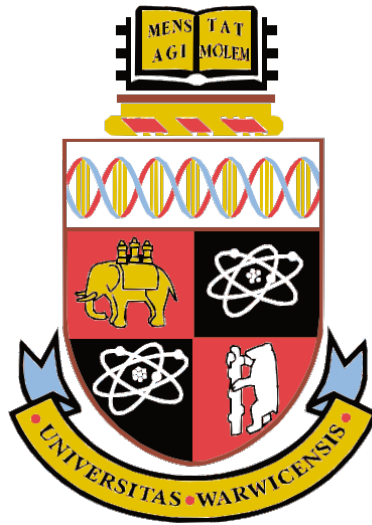




Mathematical Modelling of Microscale Evaporation using the Moment Method

by

Thomas De Fraja

MA4K9 Dissertation

Submitted to The University of Warwick

Mathematics Institute

April, 2020



Contents

1	Introduction	1
1.1	The need for a new cooling mechanism	1
1.2	The Moment Equations	2
1.2.1	The Moments	3
1.2.2	Evolution of the phase density - the Boltzmann equation	6
1.2.3	The Moment Equations	7
1.2.4	Linearised moment equations	8
1.2.5	Regularised 13 equations	9
1.2.6	Boundary Conditions	10
1.3	The Finite Element Method	12
2	Validation	13
2.1	Poiseuille flow	13
2.1.1	No slip and no temperature jump	15
2.1.2	Slip Conditions	16
2.2	Evaporation	18
2.2.1	Evaporation & condensation between liquid reservoirs	18
2.2.2	Evaporation into a half-space	21
3	Limitations	23
4	Nanoporous membrane simulations	24
4.1	Preliminaries	24
4.2	Results	25
5	The mechanics of cooling	32
6	Discussion	36
7	Conclusion	39

1 Introduction

1.1 The need for a new cooling mechanism

The development of a range of electronic devices such as laser diodes, solar cells and power amplifiers is approaching a critical bottleneck; as the power outputs increase and the sizes decrease, the issue of thermal management is tightening its grip and restricting the rate of advancement of these products. For instance Gallium Nitride (GaN) based power amplifiers generate hot spots on the micro scale with heat fluxes of over $1\text{kW}/\text{cm}^2$, with temperatures reaching over 180°C , and temperature fluctuations of around 40°C over the space of a few microns [1, p.19]. Such high temperatures are likely to lead to a decrease in efficiency and short operating lifetime.

Traditional cooling techniques involve stacking several high conductivity heat dispersers and thermal interface materials and cooling them with air cooled heat sinks or liquid cooled plates. These single-phase cooling methods have been shown to be capable of dissipating up to $1.25\text{ kW}/\text{cm}^2$ [9], however they are limited in their application as they require high operating temperatures and substantial pumping power to supply the coolant, as well as considerable sizes.

The most promising solutions involve utilising a phase-change to transfer heat away, in particular capillary pressure driven evaporation. The capillary pressure p_c calculated using the equation $p_c = \frac{2\gamma \cos \theta}{r_c}$, where r_c is the radius of the capillary, γ the surface tension of the liquid, and θ the contact angle of the liquid. New micro-structures and nano-structures known as wicks exploit the increase in capillary pressure that comes with decreasing length. A high capillary pressure means that the wicking process is essentially self-pumping; it need only rely on the capillary pressure to mediate the outgoing mass flux due to evaporation, by drawing up more water. Thus with a capillary driven flow there is no need for substantial pumping power. Although this is desirable, it leads to an increase in viscous pressure loss, with the maximum heat flux occurring when the viscous pressure loss supersedes the maximum capillary pressure. Hence early designs begin to suffer in overall performance from this if they are required to be smaller [15]. The main reason for this loss in viscous pressure is the thickness of the wick (overall thickness $600\mu\text{m}$, with height $50\mu\text{m}$ [4]). So we must go thinner.

Recent developments are utilising a nanoporous membrane device (pore diameter $\sim 100\text{nm}$), exploiting a thin membrane (thickness $< 1\mu\text{m}$) to decouple the capillary and viscous pressure [8, 5], see figure 4. This leads to much higher heat dissipating capabilities, up to $2\text{ kW}/\text{cm}^2$, even surpassing those of single-phase cooling devices [5]. This can have a significant effect on the thermal management of electronic devices such as the GaN based power amplifiers, capable of reducing maximum operating temperatures by over 40°C [8].

There are various different factors that effect the evaporation capability, including size of

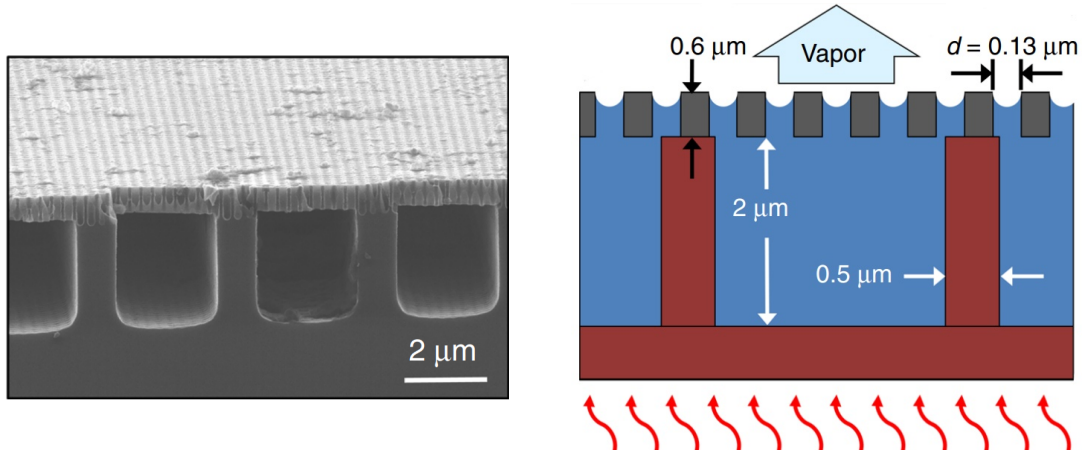


Figure 1: Left: Scanning electron microscope image of a nanoporous membrane device. Right: cross sectional schematic of the device. [5]

the nanopores, thickness of the membrane, and the liquid used to facilitate evaporation. The liquid will also have an effect on the shape of the meniscus formed inside the nanopores, which will effect the evaporative surface area, thus affecting evaporation rates. Selecting the liquid with the surface tension that gives the most effective meniscus shape will increase efficiency. Optimising the parameters of the nanoporous membrane device will increase the cooling efficiency of the device. Large parameter spaces can be challenging to explore experimentally and costly in both time and money. Computational models offer a cheaper alternative to experiments. An accurate computer model of the system could then be used to find the optimal parameters for the device. In this report we investigate several computational models and compare their accuracy.

1.2 The Moment Equations

In our day to day lives we do not need to consider the countless interactions between molecules that make up the gases around us. In fact, for many flows of material we might want to consider, we are able to obtain concrete laws governing the flow by treating the set of molecules as a continuum rather than individual points. For example, NSF laws are derived considering only conservation of mass, momentum and energy of the continuum describing the material. The Knudsen number is defined as ratio between the molecular mean free path λ , and the characteristic length scale of the system, L . The NSF laws are generally applicable for continuum flows, i.e. for $\text{Kn} < 0.01$ (roughly). However, for problems such as ours, the length scale becomes small enough so that NSF is no longer viable (for reference, at ambient pressure $\lambda_{\text{air}} \approx 68\text{nm}$, pore width $L \approx 130\text{nm}$). So we turn to new methods.

One method frequently used is the Direct Simulation Monte Carlo (DSMC) method, which simulates the individual molecules at a particular time, calculates the forces acting on them, and then advances the simulation a small time step based on these forces, then

repeats. This method is known to be very accurate for molecular flows with $\text{Kn} > 1$. The number of molecules in most systems considered however are in the order of 10^{23} , and while DSMC groups many of these particles together to make computation manageable, simulations often take a very long time to complete.

We thus seek a set of macroscopic laws to help save time with simulations. A probabilistic approach avoids the need for modelling a vast quantity of molecules, yet still retains effects from molecular interactions. The following section introduces the probability density function used to model rarefied gases, along with its moments. We will then derive an equation telling us how this function evolves, which will then give us a set of macroscopic equations for rarefied gases.

1.2.1 The Moments

The derivation of the moment equations presented below is based on that done by Struchtrup [11]. Throughout we will make use of index notation, briefly described here. A vector $\mathbf{v} = (v_1, \dots, v_n)$ can be written instead as v_i , where $i = 1, \dots, n$. When an index is repeated in the same term this signifies summing over that particular index, e.g. $\mathbf{v} \cdot \mathbf{w} = v_k w_k = v_1 w_1 + \dots + v_n w_n$, or

$$\nabla \cdot \mathbf{v} = \frac{\partial v_k}{\partial x_k} = \sum_{i=1}^n \frac{\partial v_k}{\partial x_k}.$$

This also applies to general tensors, e.g. we may have $\frac{\partial \sigma_{ik}}{\partial x_k} = \sum_{i=1}^n \frac{\partial \sigma_{ik}}{\partial x_k}$.

A gas contained in a space $V \subset \mathbb{R}^3$ consists of a set of particles α , indexed by a set $\mathcal{A}$. We will assume a monatomic gas, where each particle has a mass m . A particle will have a trajectory in space, that is, at time t the particle has position $\mathbf{x}^\alpha(t) = \{x_1^\alpha(t), x_2^\alpha(t), x_3^\alpha(t)\} \in V$ with $V \subset \mathbb{R}^3$. These trajectories have an associated velocity at the same instance in time, $\mathbf{c}^\alpha(t) = \{c_1^\alpha(t), c_2^\alpha(t), c_3^\alpha(t)\} \in \mathbb{R}^3$. If we know $(\mathbf{x}^\alpha(t), \mathbf{c}^\alpha(t)) \forall \alpha \in \mathcal{A}, \forall t \in \mathbb{R}$, then we have a complete description of the gas.

Phase density

It would not be possible to know the exact trajectories of all particles in the system, so we aim instead to describe the distribution of particles in the phase space. The phase density, $f(\mathbf{x}, \mathbf{c}, t)$, is a distribution function defined such that the number of particles at time t which are located in the subset $[\mathbf{x}, \mathbf{x} + d\mathbf{x}]$ with velocities in the range $[\mathbf{c}, \mathbf{c} + d\mathbf{c}]$, is given by

$$N_{\mathbf{x}, \mathbf{c}} = f d\mathbf{x} d\mathbf{c}. \quad (1)$$

Mass density, velocity and momentum

Integrating (1) over all possible velocities gives us the number of particles in the $d\mathbf{x}$ size box in V . If we divide by $d\mathbf{x}$ we obtain the number density (number of particles per unit volume)

$$n = \iiint f d\mathbf{c}. \quad (2)$$

For notational ease we will suppress the triple integral into one. We can then define a probability distribution for the velocity of particles, $F = \frac{f}{n}$, i.e. $\int F d\mathbf{c} = 1$. If the mass of each particle is m , then the density of the gas is

$$\rho = mn = m \int f d\mathbf{c}. \quad (3)$$

The mean velocity of the gas in the direction i is given as

$$v_i = \int c_i F d\mathbf{c} \quad (4)$$

resulting in the momentum density $\rho v_i = m \int c_i f d\mathbf{c}$.

Energy

It is often useful to shift to the frame of reference of the moving gas, so we define the peculiar velocity of a particle as $C_i = c_i - v_i$. The specific internal energy density of the system (considering only the energy density of the individual molecules, and ignoring the kinetic energy of the whole macroscopic system) is given by

$$w = \int \frac{1}{2} C^2 F d\mathbf{c},$$

with $C^2 = C_1^2 + C_2^2 + C_3^2$.

Pressure and temperature

Struchtrup [11] derives pressure as

$$p = \frac{1}{3} \rho \int C^2 F d\mathbf{c} = \frac{2}{3} mnw = \frac{2}{3} \rho w. \quad (5)$$

We will only be considering ideal gases, and the ideal gas law gives $p = \frac{k}{m} \rho T$, where k is the Boltzmann constant. We thus write the temperature as $T = \frac{2}{3} \frac{m}{k} w$. For ease of notation we instead take

$$\theta = \frac{k}{m} T = \frac{2}{3} w = \frac{1}{3} \int C^2 F d\mathbf{c}. \quad (6)$$

The ideal gas law then becomes $p = \rho \theta$.

Other moments

We continue in this fashion, multiplying the probability function F by combinations of C_i & C^{2n} to obtain more moments, few more of which have physical significance. The pressure tensor p_{ij} , stress tensor σ_{ij} and the heat flux q_i are the last such moments, and are defined as

$$p_{ij} = \rho \int C_i C_j F d\mathbf{c}, \quad \sigma_{ij} = p_{\langle ij \rangle} = p_{ij} - p \delta_{ij} \quad q_i = \frac{\rho}{2} \int C_i C^2 F d\mathbf{c}, \quad (7)$$

with $A_{\langle ij \rangle}$ denoting the symmetric trace free part of a tensor A_{ij} ($p = \text{tr}(p_{ij}) = p_{ii}$).

The above moments of F , given in (3)-(7), give us the 13 variables that appear in the moment equations, which we shall denote $v_A^{[13]}$, so that

$$v_A^{[13]} = \rho \int \Phi_A^{[13]}(c_i) F d\mathbf{c} = \{\rho, \rho v_i, \rho \theta, \sigma_{ij}, q_i\}, \quad (8)$$

after multiplication by ρ . $\Phi_A^{[13]}(c_i)$ is a vector of polynomials of the particle velocities c_i , so for the $v_A^{[13]}$ we have taken

$$\Phi_A^{[13]}(c_i) = \left\{ 1, c_i, \frac{1}{3} C^2, C_{\langle i} C_{j \rangle}, \frac{1}{2} C^2 C_i \right\}. \quad (9)$$

However, we will see that the equations themselves can contain the higher moments m_{ijk}, R_{ij}, Δ , with

$$m_{ijk} = \rho \int C_{\langle i} C_j C_{k \rangle} F d\mathbf{c}, \quad R_{ij} = \rho \int C^2 C_{\langle i} C_{j \rangle} F d\mathbf{c}, \quad \Delta = \rho \int C^4 F d\mathbf{c},$$

so the system is not closed. To close an unclosed system we must find expressions for these higher terms in terms of the 13 moments $v_A^{[13]}$, and we do this by approximating the phase density f as an expansion of the equilibrium phase density

$$f_E = \frac{\rho}{m} \frac{1}{\sqrt{2\pi\theta^3}} \exp \left[-\frac{(c_i - v_i)^2}{2\theta} \right], \quad (10)$$

i.e. we may write

$$f|_{13} = f_E \sum_A \lambda_A \Phi_A. \quad (11)$$

We must then find the coefficients λ_A such that if we take $f|_{13}$ as our phase density, it reproduces the moments $v_A^{[13]}$. Then we calculate the higher moments in terms of $f|_{13}$, rather than f .

We now have the variables which describe the gas, and so we need to know how they are all related to give us a set of macroscopic equations. In the next section we derive an

equation telling us how the phase density evolves, called the Boltzmann equation. Above, when we integrated the phase density, it gave us the constituent variable. Thus, if we integrate the Boltzmann equation, it should tell us how the moments themselves evolve.

1.2.2 Evolution of the phase density - the Boltzmann equation

We consider a fixed subset of the phase space Ω , and a infinitesimal surface element dS_Ω on the boundary of Ω , with normal n_A , where $A = 1, \dots, 6$. A point $\mathbf{p} \in \Omega$ is of the form $(x_1, x_2, x_3, c_1, c_2, c_3) =: \mathbf{p}_A$, so we introduce the phase space vector at the point $\mathbf{p}$ as $\xi_A = \{\mathbf{x}, \mathbf{c}\}_A$, and its corresponding velocity $\dot{\xi}_A = \{\dot{\mathbf{x}}, \dot{\mathbf{c}}\}_A$. In the time interval dt , all particles with velocity $\dot{\xi}_A n_A$ and occupying the subspace with volume $dt \dot{\xi}_A n_A dS_\Omega$ lying just inside the boundary of Ω , will cross the boundary of Ω , and so contribute negatively to the number of particles in Ω . By integrating over all such surface elements, i.e. over the whole boundary $\partial\Omega$, and dividing by dt , we get

$$\frac{dN_\Omega}{dt} = - \oint_{\partial\Omega} \dot{\xi}_A n_A f dS_\Omega, \quad (12)$$

where $N_\Omega = |\{\text{particles in } \Omega\}| = \iint_\Omega f d\mathbf{x} d\mathbf{c}$. Using the divergence theorem then gives

$$\frac{dN_\Omega}{dt} = - \iint_\Omega \frac{\partial \dot{\xi}_A f}{\partial p_A} d\mathbf{x} d\mathbf{c} \quad \text{i.e.} \quad \iint_\Omega \frac{\partial f}{\partial t} + \frac{\partial \dot{\xi}_A f}{\partial p_A} d\mathbf{x} d\mathbf{c} = 0.$$

Since this holds for all subsets Ω , the integrand must be zero, so we obtain that the evolution of f satisfies

$$\frac{\partial f}{\partial t} + c_k \frac{\partial f}{\partial x_k} + \frac{\partial \dot{c}_k f}{\partial c_k} = 0 \quad (13)$$

using the fact that $\frac{\partial f c_k}{\partial x_k} = f \frac{\partial c_k}{\partial x_k} + c_k \frac{\partial f}{\partial x_k} = c_k \frac{\partial f}{\partial x_k}$, since a particle's velocity is independent of its position.

From here we can split the acceleration term $\dot{c}_k$ into external forces, G_k , and intermolecular forces W_k . Thus one can rewrite (13) as

$$\frac{\partial f}{\partial t} + c_k \frac{\partial f}{\partial x_k} + G_k \frac{\partial f}{\partial c_k} = \mathcal{S}, \quad (14)$$

with $\mathcal{S} = -\frac{\partial W_k f}{\partial c_k}$, since external forces are independent of a particles velocity. An expression for the particle interaction term $\mathcal{S}$ can be found by considering two particles colliding at a point $\mathbf{x}$ in the configuration space with velocities $\mathbf{c}$ and $\mathbf{c}^1$. Then, by treating the molecules as hard spheres and considering the resulting velocities $\tilde{\mathbf{c}}, \tilde{\mathbf{c}}^1$, we can then integrate over all these possible pre- and post- velocities. This gives us the number of particles colliding, thereby either producing or destroying a phase point $\mathbf{p}$, i.e the change, or evolution, of the phase density due solely to the interaction of the particles.

With the expression for $\mathcal{S}$ in hand, the evolution of f is given by the so-called Boltzmann

equation for hard spheres, with (14) becoming

$$\frac{\partial f}{\partial t} + c_k \frac{\partial f}{\partial x_k} + G_k \frac{\partial f}{\partial c_k} = \int \int_0^{2\pi} \int_0^{\frac{\pi}{2}} (\tilde{f} \tilde{f}^1 - f f^1) g d^2 \cos \Theta \sin \Theta d\Theta d\epsilon d\mathbf{c}^1, \quad (15)$$

where we have switched to polar coordinates in the configuration space, d is the diameter of the molecules, and $g = (\mathbf{c} - \mathbf{c}^1)^2$ is the absolute relative velocity between particles. For notational ease we have simply let $f^1 = f(\mathbf{x}, \mathbf{c}^1)$ and $\tilde{f} = f(\mathbf{x}, \tilde{\mathbf{c}})$.

1.2.3 The Moment Equations

The Boltzmann equation tells us how the phase density evolves in time. As we have seen, integration of the phase density gave us the moment for density, *rho*. Integrating the Boltzmann equation 15 should then tell us how ρ evolves, i.e the constituent equation for change in mass.

Integrating the Boltzmann equation (14) over the velocity space we obtain

$$\int \frac{\partial f}{\partial t} d\mathbf{c} + \int c_k \frac{\partial f}{\partial x_k} d\mathbf{c} + \int G_k \frac{\partial f}{\partial c_k} d\mathbf{c} = \int S d\mathbf{c}. \quad (16)$$

Again we use the fact that a particle's velocity is independent of its position, so that $\frac{\partial f c_k}{\partial x_k} = c_k \frac{\partial f}{\partial x_k}$, and that (briefly reintroducing the triple integral notation)

$$\begin{aligned} \iiint G_k \frac{\partial f}{\partial c_k} d\mathbf{c} &= - \iiint f \frac{\partial G_k}{\partial c_k} d\mathbf{c} + \iint_{\Gamma} f G_k n_k dA \\ &= - \iiint f \frac{\partial G_k}{\partial c_k} d\mathbf{c} \end{aligned} \quad (17)$$

where Γ is the set of particles with infinite velocity, which is clearly empty. Equation (17) vanishes under the assumption that external forces are independent of a particles velocity.

Newtons third law states that a particle α_1 exerting a force on particle α_2 is subject to an equal and opposite force from particle α_2 , and so the right hand side of (16) also vanishes. Swapping the derivatives and integrals, we obtain

$$\frac{\partial}{\partial t} \int f d\mathbf{c} + \frac{\partial}{\partial x_k} \int c_k f d\mathbf{c} = 0. \quad (18)$$

Multiplication by m yields the familiar conservation of mass equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho v_k}{\partial x_k} = \frac{D\rho}{Dt} + \rho \frac{\partial v_k}{\partial x_k} = 0, \quad (19)$$

$\frac{D}{Dt} = \frac{\partial}{\partial t} + v_k \frac{\partial}{\partial x_k}$ is the usual material derivative.

Multiplying the Boltzmann equation by all entries of $\Phi_A^{[n]}$ and then integrating over the velocity space yields the n moment equations (above we have done so for $\Phi_1^{[13]} = 1$).

Multiplication by $\Phi_{i+1}^{[13]} = c_i$ and integrating yields the conservation of momentum equation

$$\rho \frac{Dv_i}{Dt} + \theta \frac{\partial \rho}{\partial x_i} + \rho \frac{\partial \theta}{\partial x_i} + \frac{\partial \sigma_{ik}}{\partial x_k} = \rho G_i \quad (20)$$

and multiplication by $\Phi_5^{[13]} = \frac{1}{3}C^2$ and subsequent integration gives the conservation of energy equation

$$\frac{3}{2}\rho \frac{D\theta}{Dt} + \rho\theta \frac{\partial v_k}{\partial x_k} + \frac{\partial q_k}{\partial x_k} + \sigma_{kl} \frac{\partial v_k}{\partial x_l} = 0, \quad (21)$$

Note here that we have only considered 5 moments, yet the additional 8 moments σ_{ij}, q_i have appeared, and so we do not have a closed system. A different closure of the system will yield different equations. If we take $\sigma_{ij} = 0, q_i = 0$, we recover the Euler equations. One can derive a different closure, through use of (10) and (11) with $f|_5$, which gives

$$\sigma_{ij} = -2\mu \frac{\partial v_{\langle i}}{\partial x_{j\rangle}} \quad q_i = -\frac{15}{4}\mu \frac{\partial \theta}{\partial x_i},$$

where μ is the viscosity. These are the Navier-Stokes-Fourier laws.

Repeating the process for the remaining 8 moments yields the Grad moment equations

$$\frac{D\sigma_{ij}}{Dt} + \sigma_{ij} \frac{\partial v_k}{\partial x_k} + \frac{4}{5} \frac{\partial q_{\langle i}}{\partial x_{j\rangle}} + 2p \frac{\partial v_{\langle i}}{\partial x_{j\rangle}} + 2\sigma_{k\langle i} \frac{\partial v_{j\rangle}}{\partial x_k} + \frac{\partial m_{ijk}}{\partial x_k} = -\frac{p}{\mu} \sigma_{ij}, \quad (22)$$

$$\begin{aligned} \frac{Dq_i}{Dt} + \frac{5}{2}p \frac{\partial \theta}{\partial x_i} + \frac{5}{2}\sigma_{ik} \frac{\partial \theta}{\partial x_k} + \theta \frac{\partial \sigma_{ik}}{\partial x_k} - \theta \sigma_{ik} \frac{\partial \ln \rho}{\partial x_k} + \frac{7}{5}q_k \frac{\partial v_i}{\partial x_k} + \frac{2}{5}q_k \frac{\partial v_k}{\partial x_i} \\ \frac{7}{5}q_i \frac{\partial v_k}{\partial x_k} + \frac{1}{2} \frac{\partial R_{ik}}{\partial x_k} + \frac{1}{6} \frac{\partial \Delta}{\partial x_i} + m_{ilk} \frac{\partial v_l}{\partial x_k} - \frac{\sigma_{il}}{\rho} \frac{\partial \sigma_{lk}}{\partial x_k} = -\frac{2}{3} \frac{p}{\mu} q_i. \end{aligned} \quad (23)$$

Again, to close this system, we make use of (10) and (11), but with $f|_{13}$ instead. The set of equations (19)-(23) are called the Grad13 equations, for which the higher moments are given as

$$m_{ijk} = 0, \quad R_{ij} = 0, \quad \Delta = 0.$$

1.2.4 Linearised moment equations

Equations (22) & (23) are daunting to solve. It would be prudent then to simplify these equations by assuming that all parameters are small deviations from an equilibrium state given by $\rho_0, \theta_0, (v_i)_0 = 0$, and we retain only linear terms in these deviations.

It is also convenient to introduce dimensionless variables, thus we write our parameters as

$$\begin{aligned}\rho &= \rho_0(1 + \hat{\rho}), \quad \theta = \theta_0(1 + \hat{\theta}), \quad p = \rho_0\theta_0(1 + \hat{\rho} + \hat{\theta}), \quad v_i = \sqrt{\theta_0}\hat{v}_i \\ \sigma_{ij} &= \rho_0\theta_0\hat{\sigma}_{ij}, \quad q_i = \rho_0\sqrt{\theta_0}\hat{q}_i, \quad \Delta = \rho_0\theta_0^2\hat{\Delta}, \quad R_{ij} = \rho_0\theta_0^2\hat{R}_{ij} \\ m_{ijk} &= \rho_0\sqrt{\theta_0}\hat{m}_{ijk}, \quad x_i = L\hat{x}_i, \quad t = \frac{L}{\sqrt{\theta_0}}\hat{t}, \quad \text{Kn} = \frac{\mu_0\sqrt{\theta_0}}{p_0L},\end{aligned}\tag{24}$$

Substituting (24) into (19)-(21), leads to the linearised conservation laws

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \frac{\partial v_k}{\partial x_k} &= 0 \\ \frac{\partial v_i}{\partial t} + \frac{\partial \theta}{\partial x_i} + \frac{\partial \rho}{\partial x_i} + \frac{\partial \sigma_{ik}}{\partial x_k} &= G_i, \\ \frac{3}{2} \frac{\partial \theta}{\partial t} + \frac{\partial q_k}{\partial x_k} + \frac{\partial v_k}{\partial x_k} &= 0,\end{aligned}\tag{25}$$

and into (22) and (23) gives

$$\begin{aligned}\frac{\partial \sigma_{ij}}{\partial t} + \frac{4}{5} \frac{\partial q_{\langle i}}{\partial x_{j \rangle}} + 2 \frac{\partial v_{\langle i}}{\partial x_{j \rangle}} + \frac{\partial m_{ijk}}{\partial x_k} &= -\frac{\sigma_{ij}}{\text{Kn}} \\ \frac{\partial q_i}{\partial t} + \frac{5}{2} \frac{\partial \theta}{\partial x_i} + \frac{\partial \sigma_{ik}}{\partial x_k} + \frac{1}{2} \frac{\partial R_{ik}}{\partial x_k} + \frac{1}{6} \frac{\partial \Delta}{\partial x_i} &= -\frac{2}{3} \frac{q_i}{\text{Kn}},\end{aligned}\tag{26}$$

with the Grad13 moments

$$m_{ijk} = 0, \quad R_{ij} = 0, \quad \Delta = 0.$$

From their equations, we see that the set of Grad13 moment equations is an approximation of the Boltzmann equations to first order in the Knudsen number. Moreover, they contain the NSF equations. This means they should always be as accurate as the NSF equations, however when NSF begin to fail, the Grad13 equations should still be fairly accurate.

1.2.5 Regularised 13 equations

The R13 equations are obtained via the closure of the system, derived by Struchtrup and Torrilhon[12], who make use of (10) and (11), but using the approximation for the 26 moment system $f|_{26}$. After insertion of (24) and linearisation the higher moments can be expressed in terms of the moments $v_A^{[13]}$ by

$$m_{ijk} = -2\text{Kn} \frac{\partial \sigma_{\langle ij}}{\partial x_k}, \quad R_{ij} = -\frac{24}{5} \text{Kn} \frac{\partial q_{\langle i}}{\partial x_{j \rangle}}, \quad \Delta = -12\text{Kn} \frac{\partial q_k}{\partial x_k}.$$

For stationary flows that we consider, Δ will always be zero.

The R13 equations are a second order extension of the Grad13 equations in the Knudsen

number, and so contain the Grad13 equations (taking only first order terms in the R13 equations returns the Grad13 set). Thus they should be just as accurate as the Grad13 equations when dealing with bulk behaviour, however one advantage to considering this extra set of moments is the additional description they give to Knudsen boundary layers. The Knudsen boundary layer is the region between the interface and one mean free path length away from the interface. Since our characteristic length scale is L and will be fixed at a dimensionless length of one, the Knudsen number is equal to the mean free path length. Thus second order accuracy in the Knudsen number should give an accurate description of the Knudsen boundary layers. This in turn should give a more accurate overall description of the behaviour of the gas, especially when Knudsen numbers become larger, and Knudsen boundary layer effects become more prominent.

1.2.6 Boundary Conditions

The following section gives an outline of the derivation of the R13 evaporative boundary conditions done by Struchtrup et al. [14]. To solve these equations we need to know what happens at the boundary of the given system, where the molecular interactions are different. At interfaces, we use the gas density distribution to describe the gas inside the interface. Once again, instead of working with the full distribution equation f , we approximate it using (11). We seek a full set of boundary conditions for the R13 equations (in turn giving us boundary conditions for the Grad13 equations), thus we take the 26 moment approximation instead, $f|_{26}$. It can be shown that [12]

$$f|_{26}(C_i) = \frac{\rho}{m} \frac{1}{(2\pi\theta)^{3/2}} \exp\left(-\frac{C^2}{2\theta}\right) \left(1 + \frac{v_n}{\theta} C_n + \frac{\Delta}{120\rho\theta^4} (15\theta^2 - 10\theta C^2 + C^4) + \frac{1}{5} \frac{q_k}{\rho\theta^3} C_k (C^2 - 5\theta) + \frac{\sigma_{ij}}{2\rho\theta^2} C_{\langle i} C_{j \rangle} - \frac{R_{ij}}{28\rho\theta^4} C_{\langle i} C_{j \rangle} (7\theta - C^2) + \frac{m_{ijk}}{6\rho\theta^3} C_{\langle i} C_j C_{k \rangle} \right). \quad (27)$$

The mean free path for particles constituting the liquid/solid interface is negligible compared to that of the gas, and so we model the interface material as a continuous boundary with no particle interactions. This does not mean to say that the interface does not contribute to the behaviour of the gas inside, in fact it has a meaningful effect on it. For example, a liquid interface will inevitably emit or absorb particles (evaporation/condensation) and these must be taken into account when we look to implement the above equations for the desired system.

Consider a gas sitting next to a stationary liquid interface (solid wall are treated the same but without any absorption or emission). The interface has (inward pointing) normal n_i , temperature θ_L and saturation pressure p_s . A particle in the gas incident on the liquid is either reflected back into the gas or absorbed into the liquid, and the liquid itself may emit a particle into the gas. A particle incident on the wall will be described by the usual phase distribution of the gas. Particles receding away from the wall do so in one of three

ways: they have been reflected perfectly off the boundary; they are thermally reflected off the boundary (meaning the velocity of the reflected particle is modelled by the Maxwellian (10), i.e. in equilibrium); or they have been emitted from the liquid, once again leaving in equilibrium. Particles emitted from the liquid will be described by the distribution function for the liquid, which we take as being in equilibrium, and thus has phase density $f_E(p_s, \theta_L, C)$.

We shall only consider perfect evaporation and condensation, so no particles will be reflected. With the above assumptions, we model the gas at the interface by the distribution function

$$f_{\text{int}} = \begin{cases} f|_{26}, & C_n \leq 0 \\ f_E(p_s, \theta_L, C), & C_n > 0 \end{cases} . \quad (28)$$

To simplify calculations, we have switched reference to the interface coordinate system, so $C_n = C_i n_i$ is the velocity in the direction of the normal to the tangent, and $C_{t_\alpha} = C_i t_{\alpha,i}$ is the velocity in one of the two tangential components of the tangent.

At the interface, we must have that the flux of all moments given by the interface distribution, f_{int} , is equal to the flux of all moments given by the distribution just inside the boundary, $f|_{26}$, i.e.

$$\int \Phi_A^{[26]} C_n f_{\text{int}} d\mathbf{c} = \int \Phi_A^{[26]} C_n f|_{26} d\mathbf{c},$$

with insertion of (28) yielding,

$$\int_{C_n > 0} \Phi_A^{[26]} C_n f_E d\mathbf{c} = \int_{C_n > 0} \Phi_A^{[26]} C_n f|_{26} d\mathbf{c},$$

with $\Phi_A^{[26]}$ being the extension of the vector of polynomials (9) to the 26 moment system.

After integration, we arrive at the boundary conditions for the regularised 13 moment system. Presented are only the linearised dimensionless boundary conditions, as these will be utilised to solve configurations which are considered. They are as follows:

Evaporation velocity,

$$v_n = \sqrt{\frac{2}{\pi}} \left(p_s - p + \frac{1}{2}(\theta - \theta_L) - \frac{1}{2}\sigma_{nn} + \frac{1}{28}R_{nn} \right), \quad (29)$$

slip velocity,

$$\sigma_{t_\alpha n} = -\sqrt{\frac{2}{\pi}} \left(v_{t_\alpha} + \frac{1}{5}q_{t_\alpha} + \frac{1}{2}m_{t_\alpha nn} \right), \quad (30)$$

temperature jump,

$$q_n = -\sqrt{\frac{2}{\pi}} \left(2(\theta - \theta_L) + \frac{1}{2}\sigma_{nn} + \frac{\Delta}{15} + \frac{5}{28}R_{nn} \right) - \frac{1}{2}v_n, \quad (31)$$

and boundary conditions for higher moments,

$$m_{nnn} = \sqrt{\frac{2}{\pi}} \left(\frac{2}{5}(\theta - \theta_L) - \frac{7}{5}\sigma_{nn} + \frac{1}{75}\Delta - \frac{1}{14}R_{nn} \right) - \frac{2}{5}v_n, \quad (32)$$

$$m_{t_\alpha t_\beta n} = -\sqrt{\frac{2}{\pi}} \left(\sigma_{t_\alpha t_\beta} + \frac{1}{14}R_{t_\alpha t_\beta} + \left(\frac{1}{5}(\theta - \theta_L) - \frac{1}{5}\sigma_{nn} + \frac{\Delta}{150} \right) \delta_{\alpha\beta} \right) + \frac{1}{5}v_n \delta_{\alpha\beta}, \quad (33)$$

$$R_{t_\alpha n} = \sqrt{\frac{2}{\pi}} \left(v_{t_\alpha} - \frac{11}{5}q_{t_\alpha} - \frac{1}{2}m_{t_\alpha nn} \right). \quad (34)$$

The Grad13 equations only require the first three boundary conditions (29)-(31), and as before, we set the higher moments to zero.

1.3 The Finite Element Method

Due to the nature of the geometries we wish to model, the above equations are impossible to solve analytically, and so we must tackle them numerically. The nature of the equations means they are well suited to be solved using the finite element method. The finite element method is a widely used and very powerful tool used to solve partial differential equations. It utilises the theory of Partial Differential Equations allows us to approximate a solution to a differential equation with the following method: Suppose we are seeking a function u satisfying the following condition,

$$\begin{cases} Du = f(x, y) & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases} \quad (35)$$

where D is some second order differential operator. Then multiplying this expression by a so called test function¹ v and integrating by parts leads us to the “weak formulation” of the problem. That is, we must find a u (the weak solution) such that for every test function v

$$\int_{\Omega} f v ds = - \int_{\Omega} \nabla u \cdot \nabla v ds. \quad (36)$$

The advantage of this now is that we have a first order differential equation, rather than a second order one. Still, the space of all test functions is an infinite dimensional space, therefore checking the condition (36) for all test functions is hard. We thus discretise the problem, by working instead in a finite subspace. This way, we need only check the condition against the basis elements of this subspace, and this in turn allows us to

¹All we need to know is that they are “nice” functions, which vanish on the boundary of Ω . More precisely, $H^1(\Omega)$ is the Sobolev space of functions v such that its norm satisfies $\|v\|_{L^2} < \infty$, and the norm of its weak derivative satisfies $\|Dv\|_{L^2} < \infty$. $H_0^1(\Omega)$ is then the subset of such functions with compact support in Ω .

approximate the weak solution as a linear combination of basis elements. So we are approximating the weak solution, which itself is an approximation of the actual solution to (35). The space Ω is divided into a smaller, more manageable parts Ω_i , called the finite elements. The basis elements of the subspace are a set of piecewise linear functions v_i , satisfying $v_i(\partial\Omega_i) = 0$. Thus, making the mesh finer will give us a closer approximation to the original space of test functions, and so should give a more accurate solution.

2 Validation

In this section, we validate a numerical model used to solve the moment equations. We use COMSOL Multiphysics (COMSOL AB, Sweden), a commercial software, to implement the finite element method. We compare the model to both analytic and previous numerical results in several different geometries. Analytic results are given for the R13 equations, whose solutions will contain the Grad13 solutions. Comparison between the Grad and R 13 models are given, and are compared to DSMC results [13][14]. We will build slowly, starting from simple geometries, gradually altering the geometry to get closer to the nanoporous simulation.

2.1 Poiseuille flow

We begin by considering a two dimensional Poiseuille flow in an infinitely long pipe of dimensionless width h . We assume an external force G_1 acting in the x_1 direction, and the lower and upper walls are at dimensionless temperatures θ_1 & θ_2 respectively, see figure 2. Independence of location along the pipe means that all parameters are functions of x_2 only. From the conservation of mass equation (25a), we have that the vertical component of velocity is constant, and since we have no velocity through the walls, the vertical velocity vanishes. So our 13 variables reduce to

$$\begin{aligned}\theta &= \theta(x_2), \quad p = p(x_2) \\ v_i &= \{u(x_2), 0, 0\}, \\ q_i &= \{q_1(x_2), q_2(x_2), 0\}, \\ \sigma_{ij} &= \begin{bmatrix} \sigma_{11}(x_2) & \sigma_{12}(x_2) & 0 \\ \sigma_{12}(x_2) & \sigma_{22}(x_2) & 0 \\ 0 & 0 & \sigma_{33}(x_2) \end{bmatrix}.\end{aligned}$$

The linearised R13 moment equations then simplify for this geometry. The conservation equations (25) become

$$\begin{aligned}\frac{\partial \sigma_{12}}{\partial x_2} &= G_1, \quad \frac{\partial p + \sigma_{22}}{\partial x_2} = 0, \\ \frac{\partial q_2}{\partial x_2} &= 0,\end{aligned}\tag{37}$$

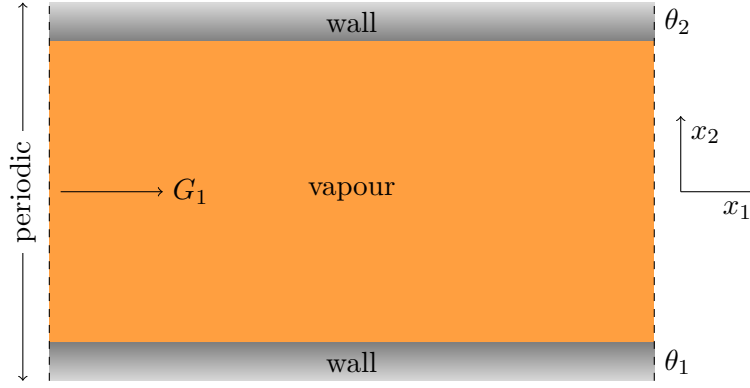


Figure 2: Setup for Poiseuille flow.

the equations for stress (26a) become

$$\begin{aligned}
 -\frac{\sigma_{11}}{\text{Kn}} &= -\frac{2}{3}\text{Kn}\frac{\partial^2\sigma_{11}}{\partial x_2^2} + \frac{4}{15}\text{Kn}\frac{\partial^2\sigma_{22}}{\partial x_2^2}, \\
 -\frac{\sigma_{22}}{\text{Kn}} &= -\frac{6}{5}\text{Kn}\frac{\partial^2\sigma_{22}}{\partial x_2^2}, \\
 -\frac{\sigma_{12}}{\text{Kn}} &= \frac{2}{5}\frac{\partial q_1}{\partial x_2} + \frac{\partial u}{\partial x_2} - \frac{16}{15}\text{Kn}\frac{\partial^2\sigma_{12}}{\partial x_2^2},
 \end{aligned} \tag{38}$$

and the equations for the heat flux (26b) become

$$\begin{aligned}
 -\frac{2}{3}\frac{q_1}{\text{Kn}} &= \frac{\partial\sigma_{12}}{\partial x_2} - \frac{6}{5}\text{Kn}\frac{\partial^2 q_1}{\partial x_2^2}, \\
 -\frac{2}{3}\frac{q_2}{\text{Kn}} &= \frac{5}{2}\frac{\partial\theta}{\partial x_2} + \frac{\partial\sigma_{22}}{\partial x_2} - \frac{24}{5}\text{Kn}\frac{\partial^2 q_2}{\partial x_2^2}.
 \end{aligned} \tag{39}$$

In (37)-(39), terms underlined denote the terms which we exclude to give us the Grad13 equations.

Solving these equations gives the general form for the variables as

$$\sigma_{12} = G_1 x_2 + \mathbf{C}_1, \quad (40a)$$

$$u = -\frac{G_1}{2\text{Kn}} x_2^2 + \mathbf{C}_1 x_2 - \frac{2}{5} q_1 + \mathbf{C}_2 \quad (40b)$$

$$q_2 = \mathbf{C}_3, \quad (40c)$$

$$\theta = \frac{15}{4} \frac{\mathbf{C}_3}{\text{Kn}} x_2 + \sigma_{22} + \mathbf{C}_4 \quad (40d)$$

$$q_1 = -\frac{3}{2} \text{Kn} G_1 + \mathbf{C}_5 \cosh\left(\frac{\sqrt{5}}{3} \frac{1}{\text{Kn}} x_2\right) + \mathbf{C}_6 \sinh\left(\frac{\sqrt{5}}{3} \frac{1}{\text{Kn}} x_2\right) \quad (40e)$$

$$\sigma_{22} = \mathbf{C}_7 \cosh\left(\sqrt{\frac{5}{6}} \frac{1}{\text{Kn}} x_2\right) + \mathbf{C}_8 \sinh\left(\sqrt{\frac{5}{6}} \frac{1}{\text{Kn}} x_2\right), \quad (40f)$$

$$\begin{aligned} \sigma_{11} = \mathbf{C}_9 \cosh\left(\sqrt{\frac{3}{2}} \frac{1}{\text{Kn}} x_2\right) + \mathbf{C}_{10} \sinh\left(\sqrt{\frac{3}{2}} \frac{1}{\text{Kn}} x_2\right) \\ - \frac{1}{2} \left(\mathbf{C}_4 \cosh\left(\sqrt{\frac{5}{6}} \frac{1}{\text{Kn}} x_2\right) + \mathbf{C}_3 \sinh\left(\sqrt{\frac{5}{6}} \frac{1}{\text{Kn}} x_2\right) \right), \end{aligned} \quad (40g)$$

Those terms which only appear in the R13 equations give rise to the hyperbolic terms, and so when considering only Grad13 equations we get that $\mathbf{C}_i = 0$ for $i = 5, \dots, 9$ in (40). Moreover, for both sets of equations and boundary conditions we will consider, the parameters q_1, u are independent of the temperatures at the wall, and so they are forced to be symmetric over the considered interval $[-h, h]$. So we may take $\mathbf{C}_1 = \mathbf{C}_5 = 0$ in (40a,b,e).

Insertion of the above into the linearised boundary conditions from the previous section allows us to set up a system of equations, leading to analytic results for both Grad13 and R13 equations.

2.1.1 No slip and no temperature jump

If the Knudsen number is small enough (< 0.02 , say) we may assume that at the walls there is no-slippage of the vapour, and there is a continuous temperature profile from the wall to the gas. We shall only consider the Grad13 equations for this regime, introducing the R13 equations when they become necessary as we move to larger Knudsen numbers.

The analytic solutions of the x -component of velocity and the temperature across the pipe

are

$$u(x_2) = \frac{G}{2\text{Kn}}(h^2 - x_2^2) \quad (41)$$

$$\theta(x_2) = \frac{\theta_2 - \theta_1}{2h}x_2 + \frac{\theta_2 + \theta_1}{2}. \quad (42)$$

This is in agreement with classical hydrodynamics, with a parabolic velocity profile and a constant temperature gradient from wall to wall.

Figure (3) shows both velocity and temperature profiles for a selected case, and overlaid are the Comsol model results, which show exact agreement with these equations.

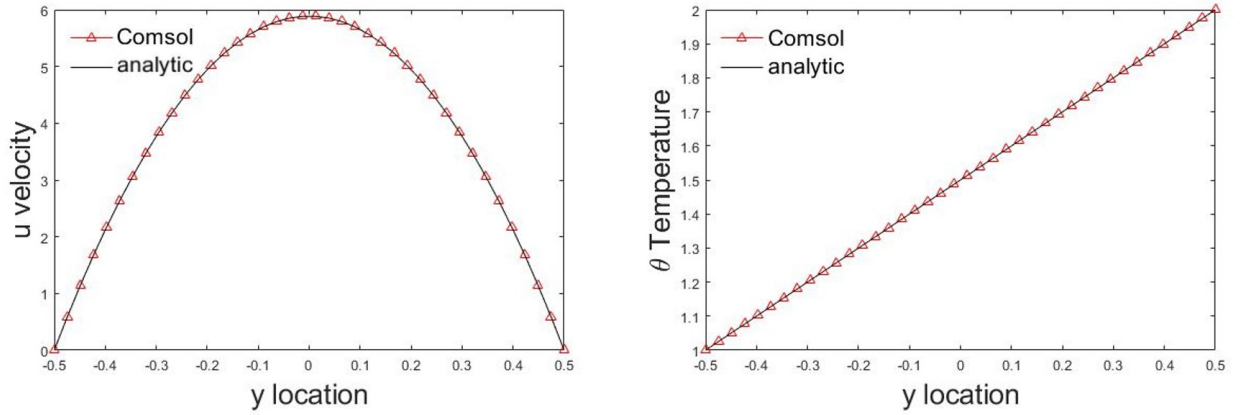


Figure 3: Velocity and Temperature profiles for Poiseuille flow for $\text{Kn}=0.005$, $G=0.2355$, $\theta_1 = 1, \theta_2 = 2$, with no-slip no-temperature jump boundary conditions. Both Comsol (solid black) and analytic expressions (red triangled) are shown.

2.1.2 Slip Conditions

With increasing Knudsen number, the boundary layer effects (and thus the boundary layer conditions) become increasingly important due to the particle-to-wall interactions becoming more prominent. Thus we introduce the linearised boundary conditions from section 1.2.6. The Grad13 system has 4 integration constants, thus needs 4 boundary conditions for a complete solution, i.e. 2 at each wall, namely (30) and (31). In a similar fashion, the R13 solution has 10 unknown constants, and so we need the full set of boundary conditions, (30)-(32), at each wall.

Solving the system for the Grad13 equations gives the solution for velocity and temperature

as

$$u(x_2) = \frac{G}{2\text{Kn}}(h^2 - x_2^2) + G \left(\sqrt{\frac{\pi}{2}}h + \frac{3}{10}\text{Kn} \right)$$

$$\theta(x_2) = \frac{\theta_2 - \theta_1}{2h} \frac{1}{\frac{1}{2} + \frac{15}{8}\sqrt{\frac{\pi}{2}}\text{Kn}} x_2 + \frac{\theta_1 + \theta_2}{2}.$$

We observe that these boundary conditions for the Grad13 equations give a bulk behaviour correction; the velocity is shifted to a higher value to accommodate for slippage, and the gradient of the temperature profile is adjusted to account for the temperature jump. The overall behaviour is not altered however, as we still have parabolic and linear profiles for the velocity and temperature, respectively.

If we look instead at the solutions to the R13 equations, we obtain the velocity profile as

$$u = \frac{G_1}{2\text{Kn}}(h^2 - x_2^2) + \left(\frac{43}{30}\text{Kn} + \sqrt{\frac{\pi}{2}} \right) G_1$$

$$+ AG_1 \left[\cosh \left(\frac{\sqrt{5}}{3} \frac{h}{\text{Kn}} \right) - 2 \cosh \left(\frac{\sqrt{5}}{3} \frac{1}{\text{Kn}} x_2 \right) \right]$$

and the temperature profile as

$$\theta = B \frac{(\theta_2 - \theta_1)}{2h} x_2 + C \sinh \left(\sqrt{\frac{5}{6}} \frac{1}{\text{Kn}} x_2 \right) + \frac{\theta_1 + \theta_2}{2}.$$

The constants A , B & C are given by.

$$A = \frac{\sqrt{\frac{\pi}{2}}h + \frac{14}{3}\text{Kn}G_1}{12 \cosh \left(\frac{\sqrt{5}}{3} \frac{h}{\text{Kn}} \right) + 20\sqrt{\frac{\pi}{10}} \sinh \left(\frac{\sqrt{5}}{3} \frac{h}{\text{Kn}} \right)},$$

$$B = \frac{8}{15} \frac{1 - \frac{10}{3}\sqrt{15\pi} \coth \left(\sqrt{\frac{5}{6}} \frac{h}{\text{Kn}} \right)}{\left(\frac{40}{3} + \frac{16}{9}\sqrt{15\pi} \right) + \left(\sqrt{\frac{\pi}{2}} + \frac{5\sqrt{30}}{3}\pi \right) \frac{\text{Kn}}{h}},$$

$$C = \frac{10}{3 \sinh \left(\sqrt{\frac{5}{6}} \frac{h}{\text{Kn}} \right)} \left((\theta_1 - \theta_2) - \left(\frac{8}{15} \frac{h}{\text{Kn}} + \sqrt{\frac{\pi}{2}} \right) \frac{15}{4} B \right).$$

We see that R13 gives slightly different bulk behaviour corrections. This stems from the addition of a hyperbolic correction term to both velocity and temperature profiles. Though the correction is small, it nonetheless reflects the fact that the R13 equations are able to capture the Knudsen boundary layer effects, unlike the Grad13 equations. We will see later that for evaporative systems the Knudsen boundary layers become more prominent in the behaviour of the gas.

Figure 4 plots the results from Comsol. We compare the Grad13 and R13 models, and

Struchtrup and Weiss' DSMC results for $Kn=0.072$ [13]. Analytic solutions are not shown, but match exactly with Comsol simulations. We see that even for relatively small Knudsen numbers such as 0.072, R13 is already showing more promising results compared to Grad13 against the DSMC data.

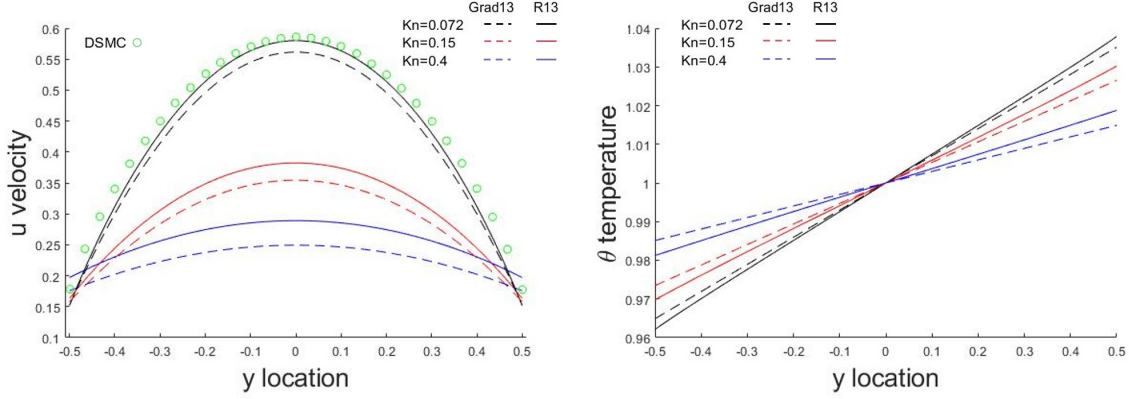


Figure 4: Velocity and Temperature profiles for Poiseuille flow for dimensionless parameters $G=0.2355$, $\theta_1 = 0.95$, $\theta_2 = 1.05$, and various Knudsen numbers.

2.2 Evaporation

2.2.1 Evaporation & condensation between liquid reservoirs

We now turn to modelling geometries involving evaporation from a liquid interface. We consider heat and mass transfer between two reservoirs of water. This introduces the evaporative boundary condition which is of great importance for the nano-porous membrane device we eventually want to consider.

The setup is as follows: we have two infinite parallel reservoirs of water separated by a dimensionless distance 1, see figure 5. We prescribe the saturation pressures of the vapour at the liquid interfaces, with the saturation pressures for the upper and lower liquid denoted p_1 and p_2 respectively. We do the same for the temperature of the liquid at each reservoir, θ_1 and θ_2 denoting the temperature for the lower and upper reservoir respectively. There are no external forces.

The linearised boundary conditions show that both pressure and temperature gradients facilitate evaporation (and subsequently condensation), and so we force evaporation at the lower interface by setting $p_1 > p_2, \theta_1 > \theta_2$. Again to simplify the problem, we only consider small deviations from an equilibrium state, thus we prescribe $p_{1,2} = 1 \pm \Delta p$ and $\theta_{1,2} = 1 \pm \Delta \theta$, and we shall only consider $\Delta p = \Delta \theta = 0.05$.

This problem can also be solved analytically. We may assume that this flow is purely one

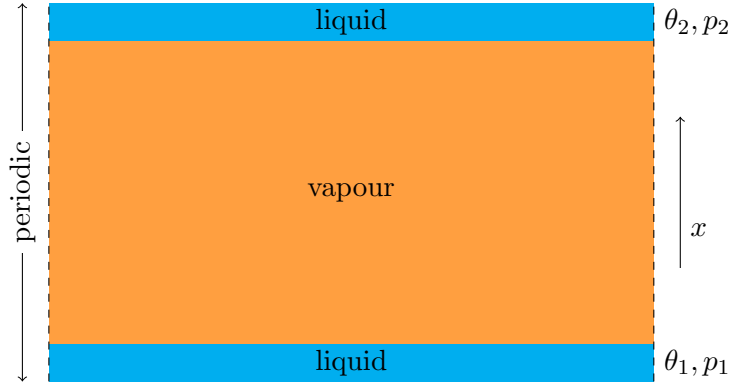


Figure 5: Setup for the problem.

dimensional, so that our variables now reduce to

$$\theta = \theta(x), \quad p = p(x), \quad v_i = u(x), \quad q_i = q(x), \quad \sigma_i = \sigma(x).$$

The one dimensional moment equations are then given by

$$\begin{aligned} \frac{\partial u}{\partial x} &= 0, & \frac{\partial p + \sigma}{\partial x} &= 0, & \frac{\partial q}{\partial x} &= 0, \\ \sigma &= \frac{6}{5} \text{Kn}^2 \frac{\partial^2 \sigma}{\partial x^2}, & -\frac{2}{3} \frac{q}{\text{Kn}} &= \frac{5}{2} \frac{\partial \theta}{\partial x} + \frac{\partial \sigma}{\partial x}, \end{aligned} \quad (43)$$

which give solutions

$$\begin{aligned} u &= u_0, \quad p + \sigma = P_0, \quad q = q_0, \\ \sigma &= \mathbf{C}_1 \sinh \left(\sqrt{\frac{5}{6\beta}} \frac{x}{\text{Kn}} \right) + \mathbf{C}_2 \cosh \left(\sqrt{\frac{5}{6\beta}} \frac{x}{\text{Kn}} \right), \\ \theta &= \mathbf{C}_3 - \frac{4q_0}{15\text{Kn}} x - \frac{2}{5} \sigma. \end{aligned} \quad (44)$$

Once again, the Grad13 moments do not contain the second order terms in (43), thus the Grad13 solution has $\mathbf{C}_1 = \mathbf{C}_2 = 0$.

Here we have six constants of integration, thus we use only 3 of the 6 boundary conditions for R13, (29), (31) and (32a), and only 2 for Grad13, (29) and (31). Insertion of (44) into these boundary conditions gives the system of equations for Grad13 as

$$\begin{aligned} u_0 &= \sqrt{\frac{2}{\pi}} \frac{1}{2} \left(p_1 - p_2 + \frac{1}{2} \left(\frac{4q_0}{15\text{Kn}} + \theta_2 - \theta_1 \right) \right), \\ q_0 &= \sqrt{\frac{2}{\pi}} \left(\frac{4q_0}{15\text{Kn}} + \theta_2 - \theta_1 \right) - \frac{u_0}{2}, \end{aligned} \quad (45)$$

and for R13 as

$$\begin{aligned}
u_0 &= \sqrt{\frac{2}{\pi}} \frac{1}{2} \left(p_1 - p_2 + \frac{1}{2} \left(\frac{4q_0}{15\text{Kn}} + \theta_2 - \theta_1 \right) - \frac{3}{5} \mathbf{C}_1 \sinh \left(\sqrt{\frac{5}{6}} \frac{1}{2\text{Kn}} \right) \right), \\
q_0 &= \sqrt{\frac{2}{\pi}} \left(\frac{4q_0}{15\text{Kn}} + \theta_2 - \theta_1 + \frac{3}{10} \mathbf{C}_1 \sinh \left(\sqrt{\frac{5}{6}} \frac{1}{2\text{Kn}} \right) \right) - \frac{u_0}{2}, \\
\sqrt{30} \mathbf{C}_1 \cosh \left(\sqrt{\frac{5}{6}} \frac{1}{2\text{Kn}} \right) &= \sqrt{\frac{2}{\pi}} \left(\frac{4q_0}{15\text{Kn}} + \theta_2 - \theta_1 + \frac{39}{5} \mathbf{C}_1 \sinh \left(\sqrt{\frac{5}{6}} \frac{1}{2\text{Kn}} \right) \right) + 2u_0,
\end{aligned} \tag{46}$$

where we have used the fact that $\sigma(-\frac{1}{2}) - \sigma(\frac{1}{2}) = 2\mathbf{C}_1 \sinh \left(\sqrt{\frac{5}{6}} \frac{1}{2\text{Kn}} \right)$. Solving for the constants u_0, q_0 in (45) and for the additional constant $\mathbf{C}_1$ as well as u_0, q_0 in (46) gives the analytic solution for both sets of moment equations.

We look at one particular case of Knudsen number, $\text{Kn}=0.235$, and make comparisons between the two moment equation models, and a DSMC model for the same problem. The DSMC data is from Struchtrup et al. [14]. Once again the analytic solution matches exactly with that of Comsol's numerical solution, and so have been omitted from the plots. All simulations are done in a two dimensional setting.

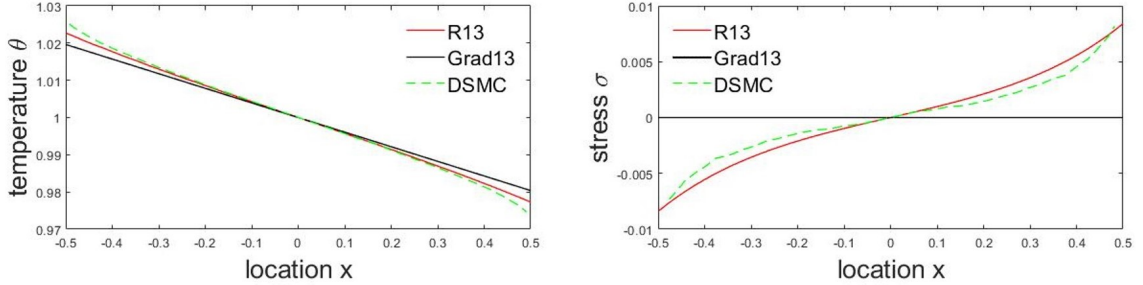


Figure 6: Temperature and stress profiles for $\text{Kn} = 0.235$, $\Delta p = \Delta \theta = 0.05$.

As can be seen in Figure 6, the temperature is accurately described by both Grad13 and R13 models, with both capturing the temperature gradient reasonably. The limitation of the Grad13 equations is however highlighted when we look at the stress, with its inability to capture Knudsen boundary layer effects, resulting in no stress throughout the gas. The R13 equations are able to accurately reproduce the stress profile due to the addition of the extra moment m_{ijk} , and its corresponding boundary condition. This also gives more accurate temperature profiles; the Grad13 model only being capable of producing a linear gradient, unlike the R13 model. This leads to discrepancies in the magnitude of the temperature jump, with the R13 vapour temperature at the interface accurate to roughly 10% compared to the DSMC simulation, whereas Grad13 is accurate to only to 26%.

We also look at the effect of varying the Knudsen number on the velocity and heat flux against DSMC results, shown in Figure 7.

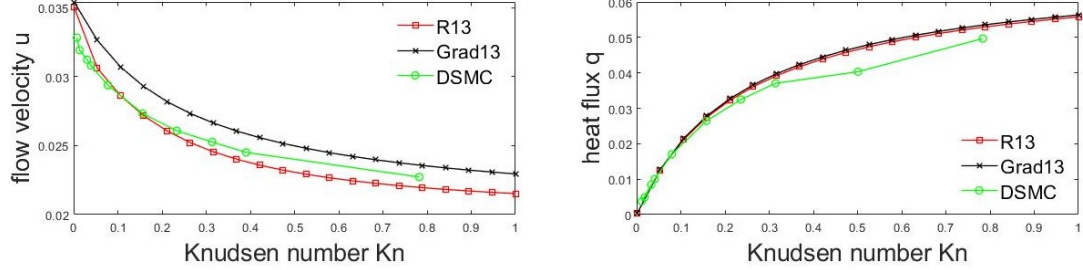


Figure 7: The effect of the Knudsen number on velocity u and heat flux q , for $\Delta p = \Delta\theta = 0.05$. Only computed values are marked.

As seen here in Figure 7, both Grad13 and R13 models in comsol are shown to be accurate at reproducing velocity and heat flow constants for various Knudsen numbers, particularly at lower Knudsen numbers. It is worth noting, as Struchtrup et al. did [14], that there may be fluctuations in the DSMC values due to the small values of u and q .

2.2.2 Evaporation into a half-space

Next we look to test the accuracy of our Comsol model for far-field simulations. This represents the assumption that far enough away from the interface, the vapour is near enough to equilibrium conditions, as would be the case for a nano-porous membrane device.

We have a similar set up as in the previous example, except at the upper boundary we assume the vapour is in a drifting equilibrium, with velocity u_∞ , sometimes called the free-stream velocity. We thus set q_∞, σ_∞ , the values of heat flux and stress at the far-field, to zero. We may prescribe one of the remaining 3 parameters $u_\infty, p_\infty, \theta_\infty$, and we shall prescribe u_∞ . We also take the saturation pressure of the liquid as $p_s = 1$, and its temperature as $\theta_L = 1$.

Once again we are able to solve this analytically. Integration of (43) with the above conditions yields

$$\begin{aligned} u &= u_\infty, & p + \sigma &= p_\infty, & q &= 0, \\ \sigma &= \mathbf{C} \exp\left(\sqrt{\frac{5}{6}} \frac{x}{\text{Kn}}\right), & \theta &= \theta_\infty - \frac{2}{5}\sigma. \end{aligned} \quad (47)$$

where once again $\mathbf{C} = 0$ gives the Grad13 expressions.

The analytic expression for the Grad13 equations is then given by inverting the following

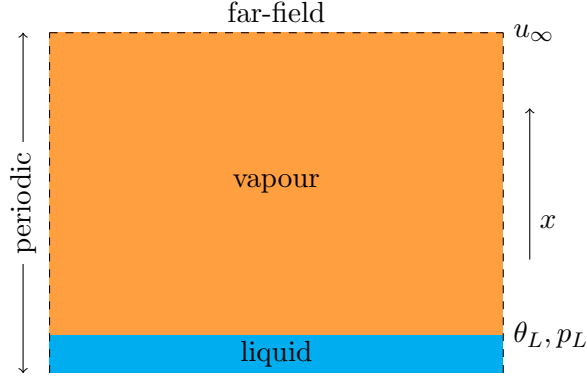


Figure 8: Setup for half-space evaporation, with velocity prescribed at the far field.

system, obtained by inserting (47) into the macroscopic boundary conditions,

$$u_\infty = \sqrt{\frac{2}{\pi}} \left(p_s - p_\infty + \frac{1}{2}(\theta_\infty - \theta_L) \right),$$

$$u_\infty = 4\sqrt{\frac{2}{\pi}} (\theta_L - \theta_\infty),$$

and the R13 expression is instead given by inverting

$$u_\infty = \sqrt{\frac{2}{\pi}} \left(p_s - p_\infty + \frac{1}{2}(\theta_\infty - \theta_L) + \frac{3}{10}\mathbf{C} \right),$$

$$u_\infty = \sqrt{\frac{2}{\pi}} \left(4(\theta_L - \theta_\infty) + \frac{3}{5}\mathbf{C} \right),$$

$$u_\infty = -\sqrt{\frac{2}{\pi}} \left(\theta_L - \theta_\infty + \frac{39}{10}\mathbf{C} \right) - \sqrt{\frac{15}{2}}\mathbf{C}.$$

The mass flux J is the mass of vapour passing through one unit area, so is given by $J = u\rho$. The analytic expression for free stream mass flux for the Grad13 solutions is

$$J_1 = u_\infty \rho_\infty = u_\infty \frac{p_\infty}{T_\infty} = u_\infty \frac{p_s - \frac{9}{8}\sqrt{\frac{\pi}{2}}u_\infty}{\theta_L - \frac{1}{4}\sqrt{\frac{\pi}{2}}u_\infty}. \quad (48)$$

Once again, our Comsol model matches precisely with the analytic solutions given by the above, and so in figure 9 we look at the comparison between the two models.

As we have seen before, the R13 set of equations allow for Knudsen layer effects, which slightly effect the far field parameters. Not only do they give different profile shapes, but the actual temperature and pressure jumps at the boundary are slightly affected.

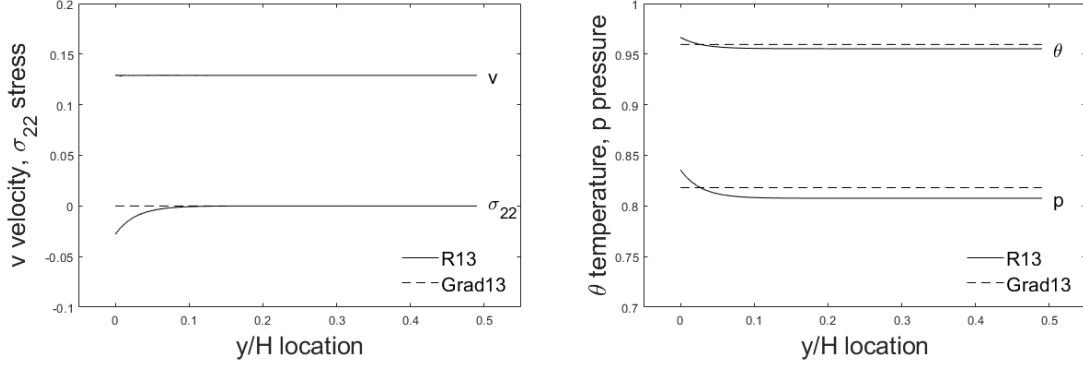


Figure 9: Far-field simulation for $u_\infty = 0.129$, and $p_s = \theta_L = 1$. The different models give similar results, with differences coming due to boundary layer effects.

3 Limitations

We already know that the moment equations -depending on how many moments are taken- are only suitable for a particular range of rarefied flows, i.e. range of Knudsen numbers. Indeed this is not a problem, as we are looking for an accurate set of equations to describe this precise regime; that is, when NSF begins to fail and we look for a less time consuming approach than DSMC (DSMC is also not necessarily 100% accurate in this regime). However, the assumptions made in section 1.2 to simplify the equations introduce other limitations for the model.

The Mach number is defined to be the ratio between the velocity and the local speed of sound, so $Ma_\infty = u_\infty / \sqrt{\gamma \frac{k}{m} T_\infty} = u_\infty / \sqrt{\gamma \theta_\infty}$. Here γ is the specific heat ratio for a monatomic gas, $\gamma = \frac{5}{3}$. We assumed, when linearising the moment equations, that the velocity only took small deviations from an equilibrium. This corresponds to $Ma_\infty \ll 1$. For larger Mach numbers however, this assumption fails, as the velocity is no longer small. Moreover, we also assumed an ideal gas, so that $p = \rho\theta$, and again this fails as the gas' velocity nears the local speed of sound.

In the far-field problem from the previous section, we look at how the free stream mass flux $J = u_\infty \rho_\infty$ is affected by the free-stream Mach number. Figure 10 highlights this limitation for higher Mach numbers. The plots for Grad13, R13 and DSMC are shown of the computed far-field mass flux J values for varying free-stream Mach numbers. The mass flux has been normalised with respect to the total evaporative mass flux $M_e = \sqrt{\frac{\theta_L}{2\pi}}$. As can be seen, even for Mach numbers as small as 0.2, we begin to see deviations from the two different methods. This is not particularly unsurprising however, as a Mach number of 0.1 corresponds to 1/10th the speed of sound, which is still something we can consider fairly speedy.

Therefore, we will only consider regimes where the Mach number is less than 0.2, where the linearised Grad 13 and R13 equations are valid and accurate.

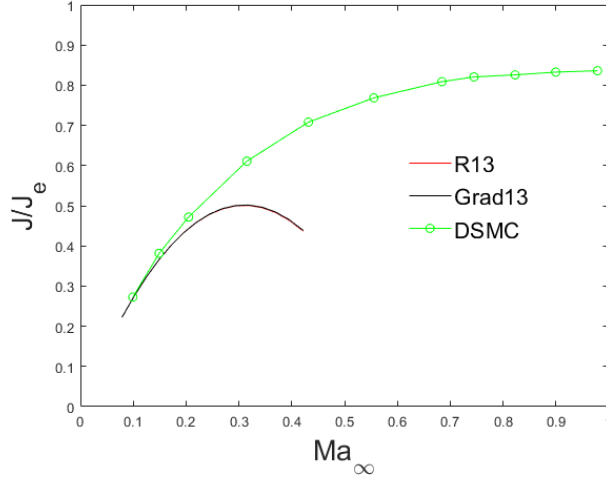


Figure 10: Far-field mass flux J , against the far-field Mach number. Comparison made between Grad and R 13 Comsol simulations, and DSMC data [6]. Our linearisation assumption breaks down for high Mach numbers.

4 Nanoporous membrane simulations

4.1 Preliminaries

We now look to model the nanoporous membranes proposed seen in figure ???. The geometry for the model is shown in figure 11. We model a single two dimensional nanopore with pore diameter L and wall width W . A large portion of the two dimensional nanoporous membrane is simulated by introducing periodic boundary conditions along the side, to allow us to greatly reduce computation time. We then fix the meniscus shape $\mathcal{M}$ of the liquid and simulate evaporation from the interface into the far-field. At the liquid interface we impose a constant saturation pressure p_s , and at both the wall and the liquid we impose a constant temperature θ_L . Boundary conditions for wall/liquid to gas interactions are derived in section 1.2.6. Once again, we prescribe the free-stream velocity u_∞ at the far-field.

The width of the walls is one of the main elements of the geometry we would like to study. The porosity is defined as the ratio between the pore width and the total width of the geometry, $\phi = \frac{L}{2W+L}$, and the wall width is changed by fixing the pore width at the characteristic length L , and varying the porosity.

Much of the work done by John et al. is in analysing the effect of varying either the Knudsen number or the Mach number [6]. DSMC begins to become less effective when large flows with small Mach numbers are considered. For instance, shock-waves can form in DSMC models when we would not expect for $Kn < 0.01$ [2], and stochastic noise becomes significant in computational models for $Ma < 0.3$ [10]. We wish to bridge the gap between regimes in which DSMC is proven to be accurate, and regimes in which we know the

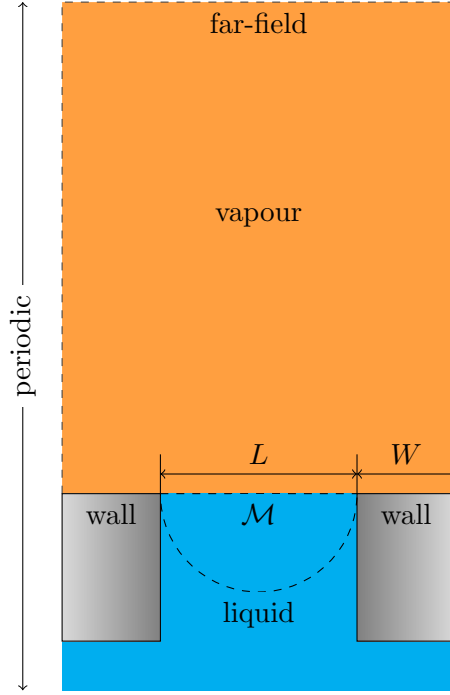


Figure 11: Geometry of a single nanopore, where the meniscus shape $\mathcal{M}$ is chosen before hand and fixed during the simulation

moment equations are accurate. We therefore consider $Ma_\infty < 0.2$ and $Kn < 0.1$.

Comparison is made between the Grad13 model, the R13 model, and DSMC results, but included are also the results from a NSF model. The NSF model will not utilise the full linearised boundary conditions as the higher moment equations have done, and we instead assume a continuous temperature profile across the interface and no slip velocity. This is the condition that $\theta - \theta_L = 0$ and $v_t = 0$ in the boundary condition equations in section 1.2.6. For most regimes in which NSF is used these conditions are not essential.

We assess the free-stream parameters J, θ_∞ and ρ_∞ , and asses their change with a varying free-stream Mach number. The two extremes for meniscus shapes are the flat meniscus and the semicircular meniscus, both of which are modelled here, and compared against each other.

4.2 Results

We begin by looking at the extreme case for the geometry. In an ideal world we would have porosity $\phi = 1$, corresponding to the membrane with infinitely thin walls, giving maximum liquid interface area for the most efficient evaporation. In reality however this is obviously infeasible, but we still consider this case nonetheless as it gives benchmark results to compare to the more realistic cases.

Figure 12 shows the results of all computed free stream parameters, for $Kn = 0.05$. The

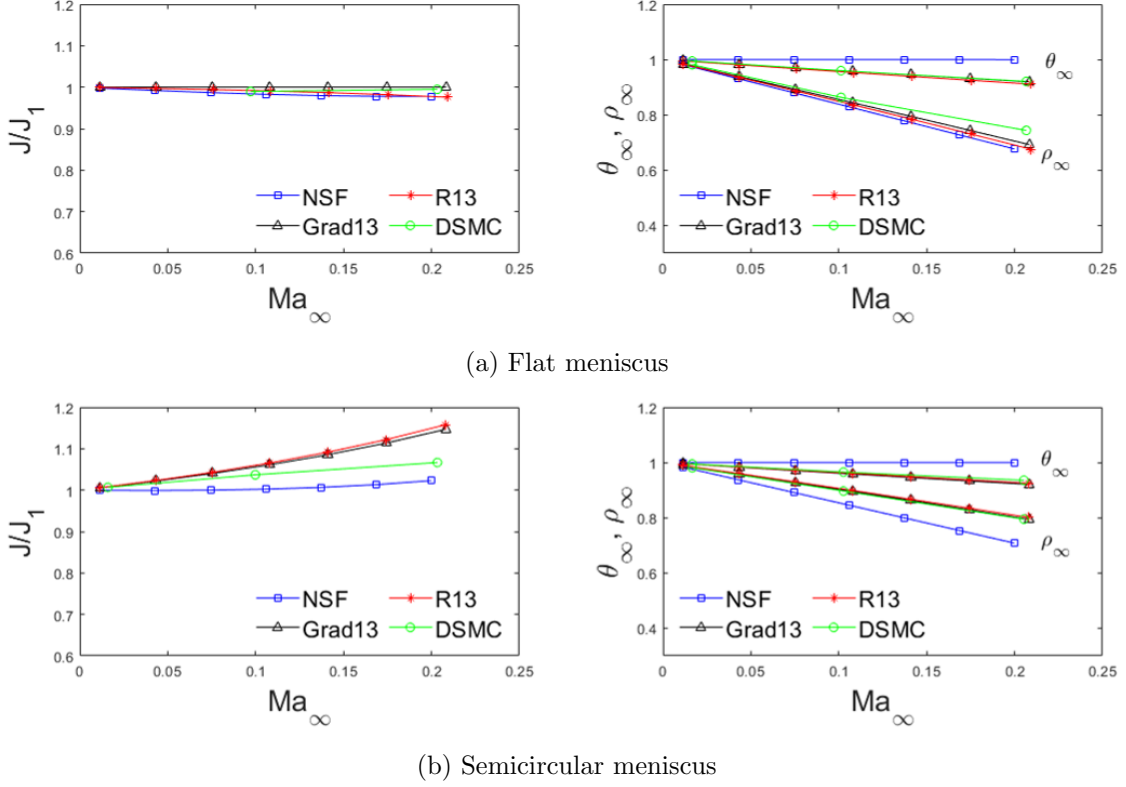


Figure 12: $\phi = 1$, the ideal case with no walls, for Knudsen number $\text{Kn} = 0.05$, and both meniscus shapes.

flat meniscus case with no walls corresponds to an infinite plane of water evaporating into a far-field, identical to the situation in section 2.2.2, and so, as expected, the ratio is always close to unity. With higher surface area for evaporation resulting from the curved meniscus, a higher mass flux is expected, and is indeed seen.

We again highlight the necessity of only looking at small Mach numbers, with computed values quickly diverging from DSMC results. Take the Grad13 model for example: if we look at the Mass flux J for the semicircular meniscus case, at $Ma_\infty = 0.1$ the Grad13 model differs from the DSMC model by just 1.7%; whereas at $Ma_\infty = 0.2$ this difference increases to 6.7%.

The continuous temperature profile at the interface we imposed for the NSF model leads to very little variation in the free-stream temperature value. This emphasizes the need for the temperature jump condition for large Knudsen numbers. The linear relation between temperature and Mach number that the DSMC results have is replicated by the moment equations, with roughly the same gradient. All macroscopic models also have decent density profiles, predicting a linear decrease in density with increasing Mach number, which for small Mach numbers seems like a good approximation of the DSMC results.

We then move on to more a realistic case, with $\phi = 0.5$, for $\text{Kn} = 0.05$. We once again consider the two cases of a flat and semicircular meniscus, and mass flux is normalised with respect to the same one dimensional case as before. Figure 13 shows the computed far field parameters for the Grad13 model compared to DSMC results. As expected, the evaporative mass flux computed for this porosity value is lower than the case with unitary porosity, as the total area available for evaporation is reduced.

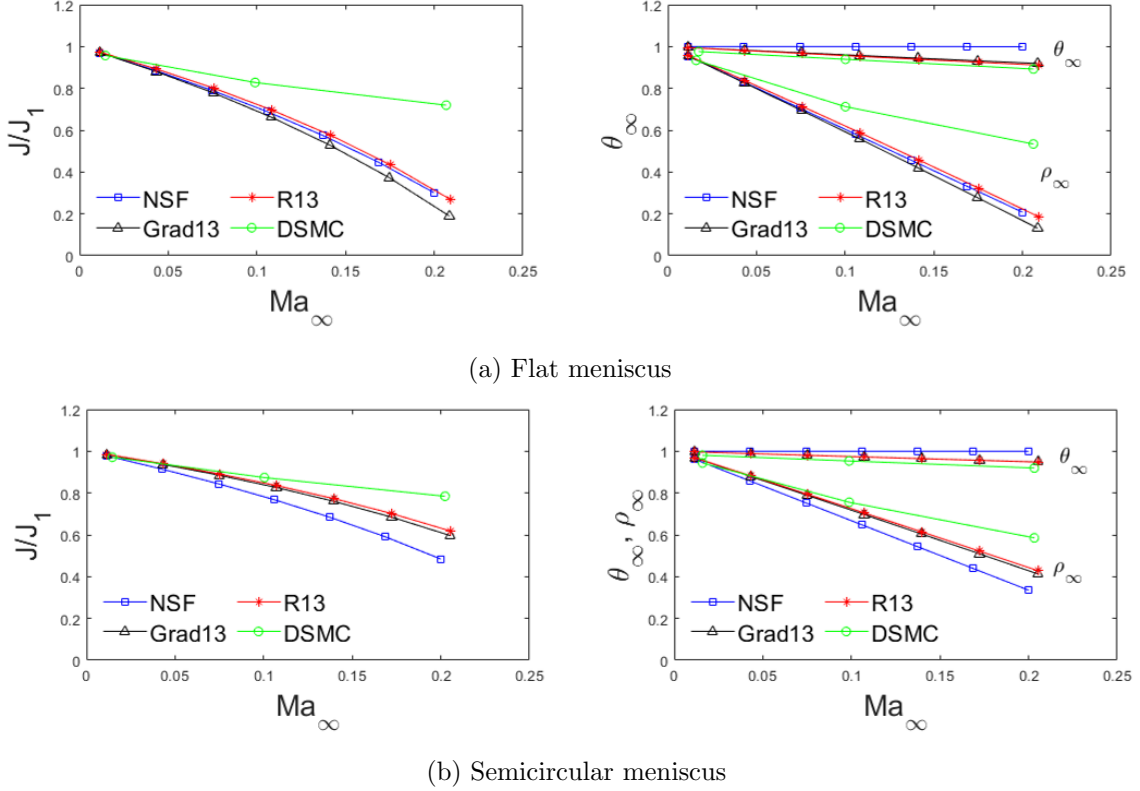


Figure 13: $\phi = 0.05$, for Knudsen number $\text{Kn} = 0.05$, and both meniscus shapes

Similar trends are seen here for $\phi = 0.5$ as with the $\phi = 1$ case. The temperature and density profiles give good approximations for the DSMC data at low Mach numbers, while the mass flux gives an average approximation for $Ma_\infty < 0.1$, but diverges from DSMC data substantially for larger Ma_∞ .

Interestingly, the 13 moment equations show improved results with the semi-circular meniscus case. This improvement is seen in both cases of porosity. For $\phi = 0.5$ and $Ma_\infty = 0.1$, the relative difference from the DSMC value in mass flux number J for the flat meniscus case is 14%, 16% and 12%, for the NSF, Grad13 and R13 models respectively. However, for the semicircular meniscus case the error in the mass fluxes are 10%, 4% and 2.9%, for the NSF, Grad13 and R13 models respectively.

Next, we shall study in-depth a particular configuration, with $\text{Kn}=0.05$, $\phi = 0.5$, and for

the semi-circular meniscus case. The surface plots for all four models and each of the three far-field parameters J , ρ_∞ and θ_∞ , are shown in figures 14-16. The results from the macroscopic equations experience the same phenomena as the DSMC model. In particular, they all exhibit a flow expansion phenomena, so that when the flow velocity is increased, the density is reduced. The R13 also exhibits this inverse relation between mass flux and temperature, similar to the DSMC results.

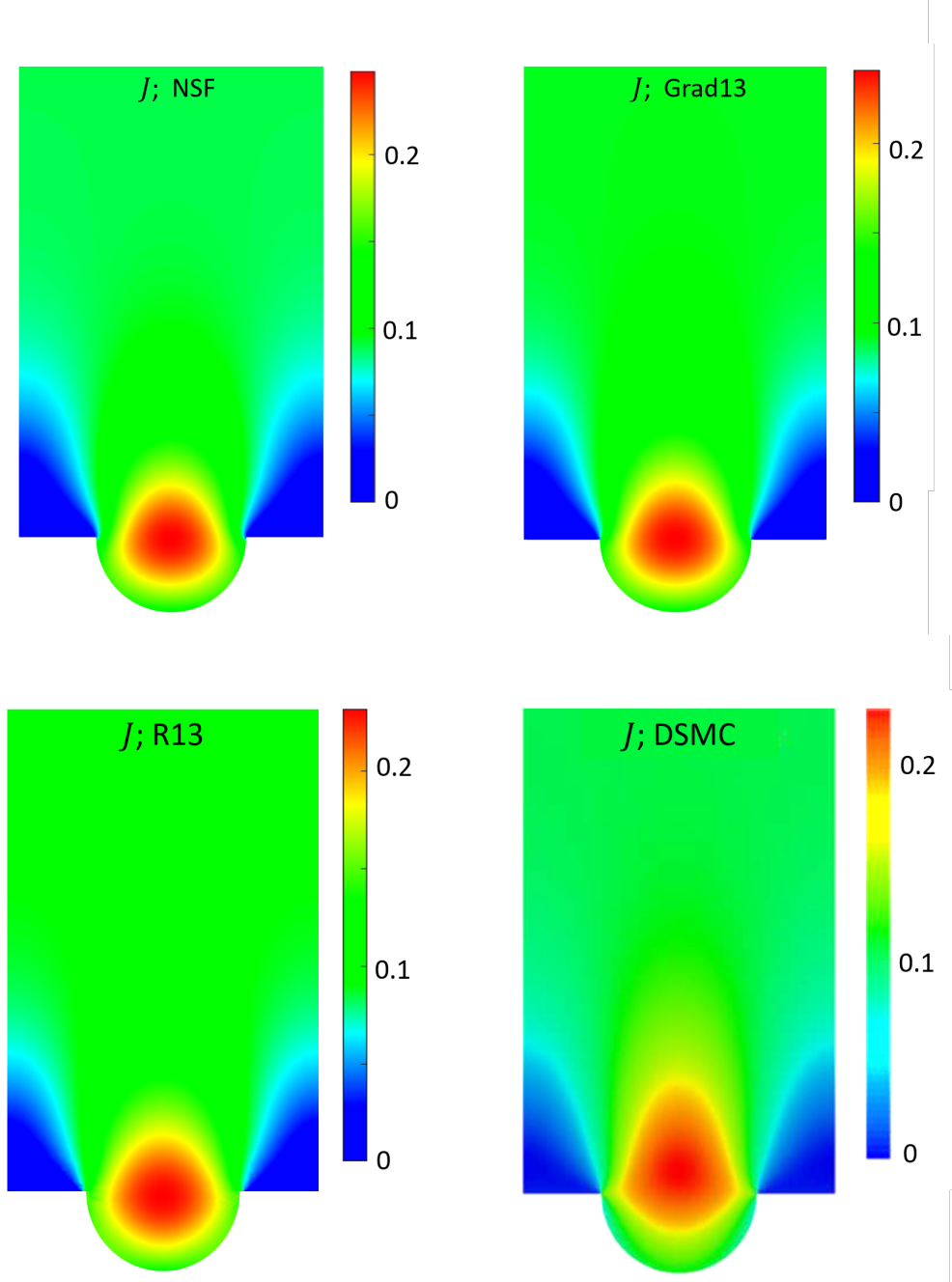


Figure 14: Mass flux surface for the semicircular meniscus case, $\phi = 0.5$, $\text{Kn} = 0.05$, free-stream velocity $U_\infty = 30\text{m/s}$. Good agreement is shown with the shapes for all models, and the values for the R13 model almost match perfectly with DSMC [6]

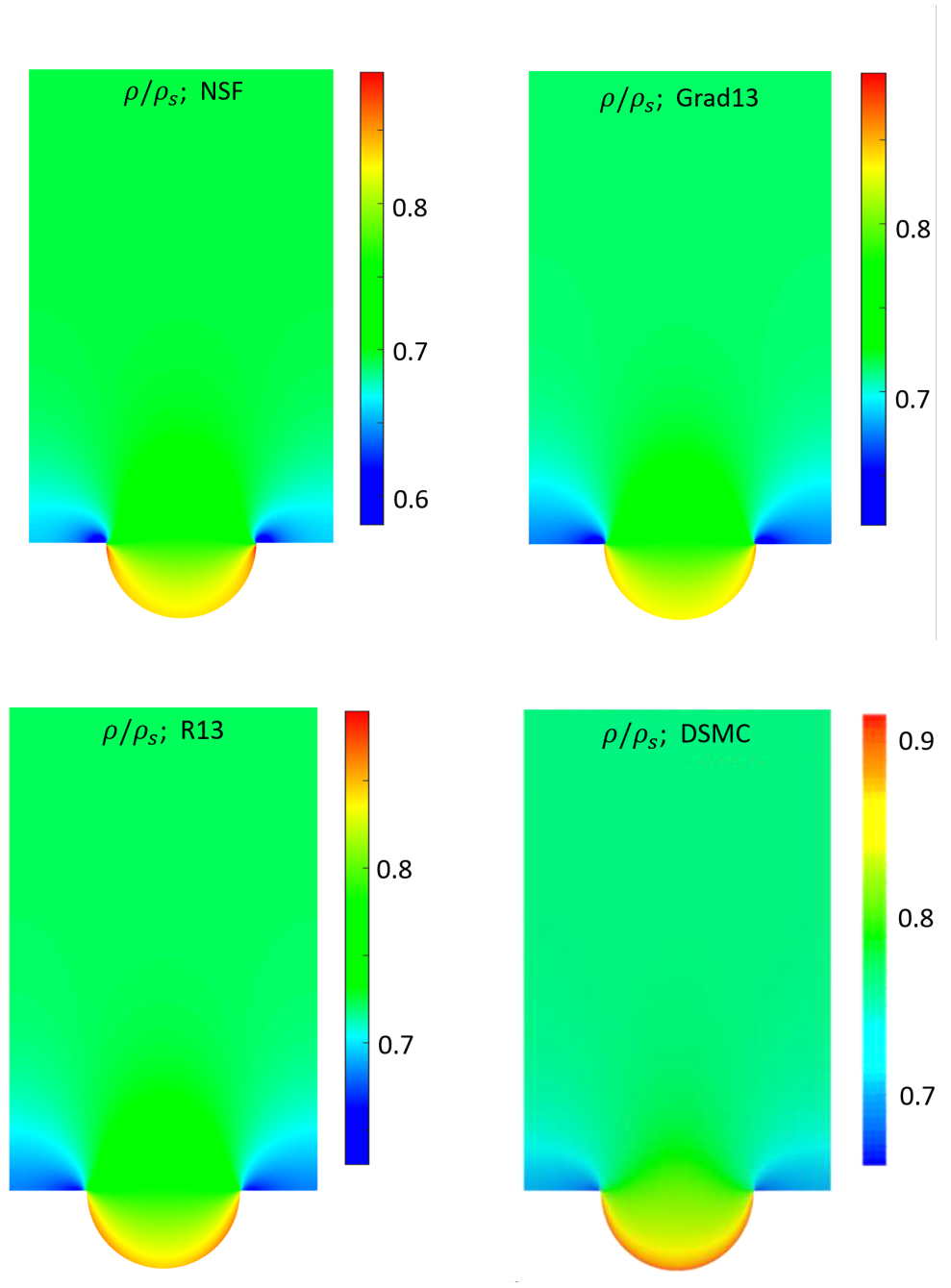


Figure 15: Density surface for the semicircular meniscus case, $\phi = 0.5$, $\text{Kn} = 0.05$, free-stream velocity $U_\infty = 30\text{m/s}$. Good agreement is shown with the shapes for all models

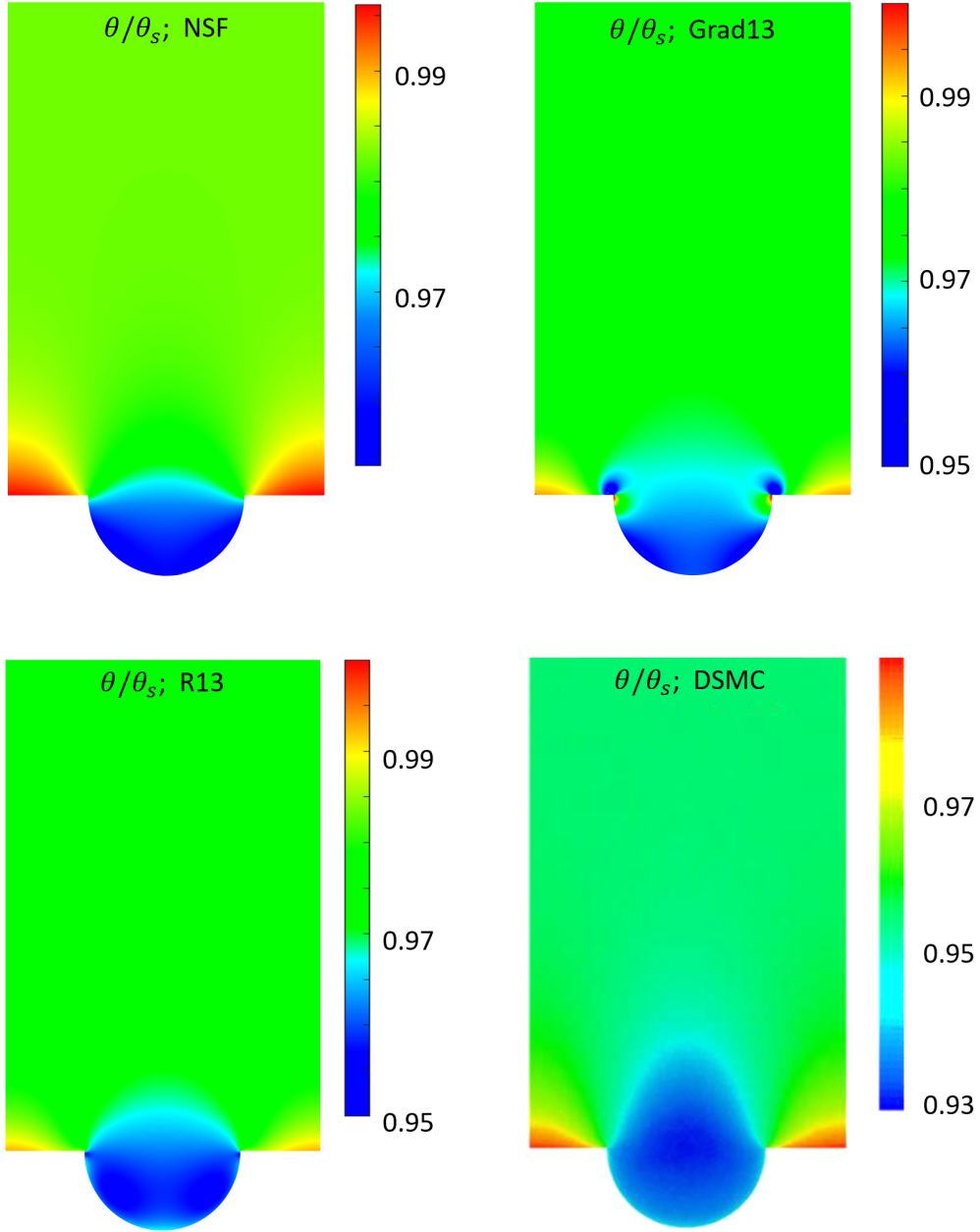


Figure 16: Temperature surface for the semicircular meniscus case, $\phi = 0.5$, $\text{Kn} = 0.05$, free-stream velocity $U_\infty = 30\text{m/s}$. Comsol struggled to resolve the stresses at the corner where the boundary conditions change, causing seemingly unnatural temperature shapes near the interfaces. This anomaly, however, seems to be localised, and so does not affect the far field parameters. Ignoring singularities, the overall shape is mostly accurate for the Grad and R 13 model, but the NSF model is seemingly less accurate.

The profiles for the flow parameters along the central vertical axis are plotted in figure 17. In section 1.2 we said that the R13 equations add a Knudsen layer correction to the Grad13 solutions, usually adding a correction on the order of the mean free path, which is the size of the Knudsen boundary layer. This was seen in 2 in the analytic solutions. We see it again in figure 17. Here, for $Kn=0.05$, since the characteristic length scale is 1, the dimensionless mean free path length is also 0.05. We see at dimensionless length $y > 0.05$, all profiles exhibit mostly the same behaviour. However, for $y < 0.05$, the parameters exhibit different behaviour, with the R13 model accurately reproducing the profile shapes. This Knudsen layer correction is most prominent in the temperature, but the density also shows behaviour of a similar vein.

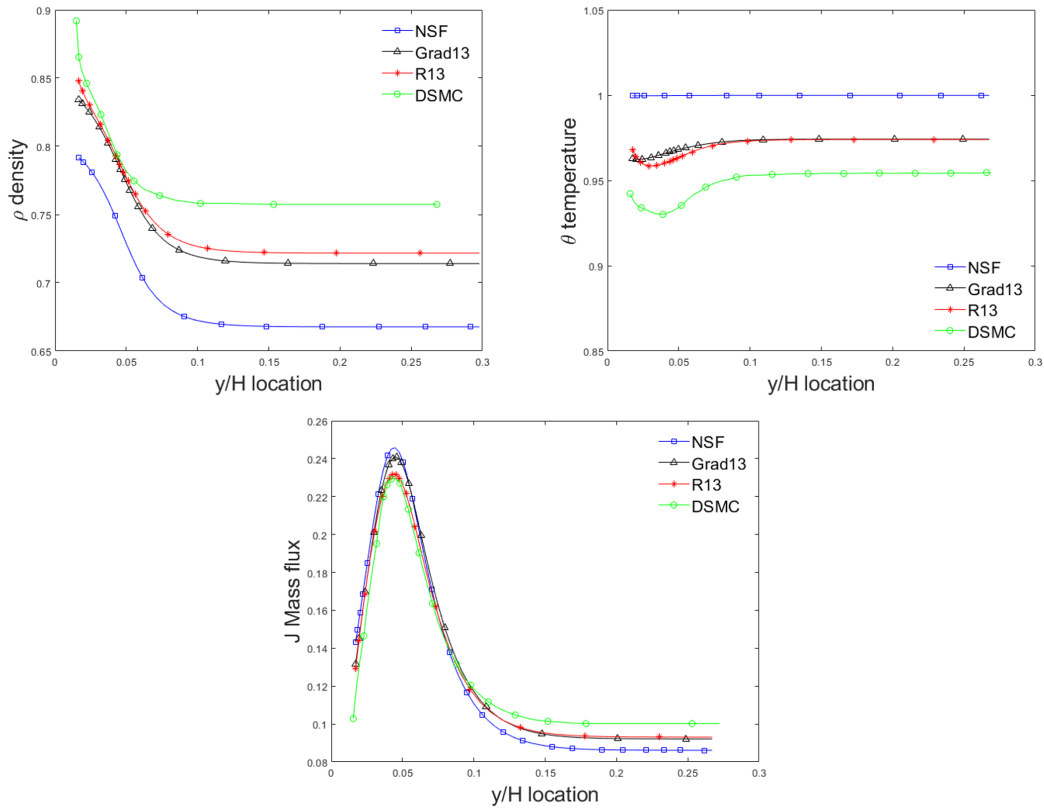


Figure 17: Flow parameters along the central vertical axis, for the semicircular meniscus case, with $Kn = 0.05$, $\phi = 0.5$ and $U_\infty = 30\text{m/s}$, corresponding to a dimensionless velocity of $u_\infty = 0.129$.

To conclude this section, we remind the reader that the goal was to accurately reproduce DSMC data using the moment equations, for specific regimes. In particular, the flow regimes with $Ma_\infty < 0.2$ and $Kn < 0.1$, being the regimes in which DSMC models become less reliable. Accurate results were indeed shown in these regimes, especially for $Ma_\infty < 0.1$. Moment equation models could also be used at very small Knudsen and Mach numbers. So, in conjunction with the DSMC data, we can give the full range of

results for a lot regimes, from very small Knudsen and Mach numbers, to very large.

5 The mechanics of cooling

One of the advantages to using a set of macroscopic equations rather than a microscopic model is that it is possible to consider systems with a wide variety of mean free path lengths, without needing additional computing resources. For the geometry in figure 11 it would be near impossible to model solid, liquid and gas at the same time in a DSMC model, due to the computational expense needed to simulate such differing mean free path lengths. However, using the finite element method, we are able to solve this problem by using an appropriate set of equations on each component to overcome the differing intermolecular distances in each material. For instance, the purpose of these nano-porous membrane devices is to cool devices down to ensure they are working efficiently. Thus far we have seen that we can describe the vapour region using the moment equations. We could, therefore, implement the heat equation on the thus far unmodelled portion of the geometry, to see how well the heat is transferred away from the device for different initial parameters.

For detailed derivations of the following equations, see Dawkins' online notes [3]. There he derives the formulas in a one dimensional setting, so here they are generalised to 3 dimensions.

Suppose we have a material occupying a region $\Omega \in \mathbb{R}^3$. Its thermal diffusivity k measures the rate of transfer of heat through the material. The heat equation then tells us that the evolution of heat $\omega(\mathbf{x}, t)$ in the region Ω is given by

$$\frac{\partial \omega}{\partial t} - k \nabla^2 \omega = 0. \quad (49)$$

Hanks et al.'s nano-porous membrane device is constructed from Silicon, and the fluid used is pentane [5]. The thermal diffusivity is known for both these materials, with $k_{\text{Si}} \approx 1000 k_{\text{pent}}$. We may assume that k is constant for both materials, since variations in temperature aren't large enough to make significant changes in material properties.

Thus we need only the boundary conditions to implement (49) into our geometry. In the above equation, we have already made use of Fourier's law², which gives the heat flux through the material as

$$q_n = -K_0 \mathbf{n} \cdot \nabla \omega, \quad (50)$$

where K_0 is the thermal conductivity. The thermal diffusivity and conductivity are related by the density ρ and specific heat c , with $k = \frac{K_0}{c\rho}$. In reality, these parameters are all dependant on $\mathbf{x}$ and t , but here we shall assume them to be constant. The moment equations have given us an expression for the heat flux at the liquid interface, and since

²the use of it can be seen in Dawkins' notes

heat flux is the heat energy transfer per unit area, the heat flux must have come from the evaporation of liquid. Therefore, at the liquid interface we impose (50). Additionally, at all interfaces we use Newton's law of cooling, which states that the heat flux at the boundary is proportional to the difference in temperature between the two materials. Thus

$$K_0 \mathbf{n} \cdot \nabla w = -H [w_b(t) - T(t)] - q_n(t) \quad (51)$$

at the liquid interface, and

$$K_0 \mathbf{n} \cdot \nabla w = -H (w_b(t) - T(t)) \quad (52)$$

for the remaining interfaces (including at the heat source). Here, w_b is the temperature at the boundary, and $T(t)$ is the temperature of the surrounding material. The constant of proportionality H must be found experimentally, and so here has been chosen to show general trends in behaviour, rather than to give accurate values of cooling.

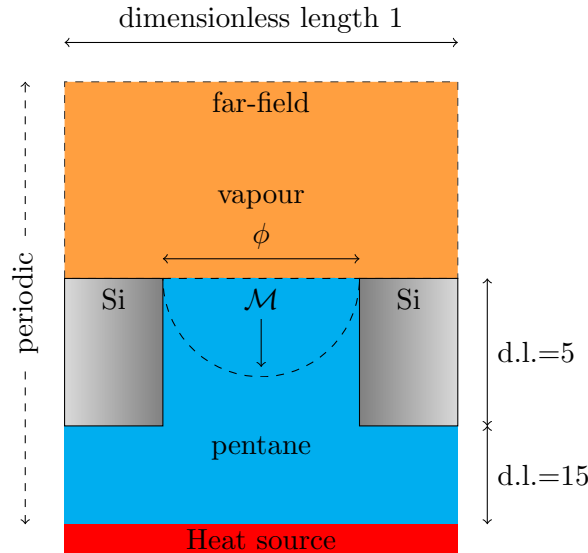


Figure 18: Revised geometry of a single nano-pore, based on the design of Hanks et al. [5]. Not to scale.

The geometry is now altered so that the characteristic length defines the length of the pore and wall together, see figure 18. The pore width is then equal to the porosity, so that we may observe the effect of porosity on the cooling of the device. We also look at the effect of the meniscus shape $\mathcal{M}$, and so we define the meniscus shape as the arc C_ϕ with radius R_ϕ going from one wall to the other³. So the semi-circular meniscus corresponds to $R_\phi = 0.5$, and the flat meniscus corresponds to $R_\phi = \infty$. $\mathcal{M}$ is therefore characterised solely by ϕ and R_ϕ .

³more specifically, the arc has radius ϕR_ϕ , and the angle swept out by the arc is $2 \arcsin\left(\frac{1}{2R}\right)$, such that the corners of the arc touch the corners of the wall.

While the heat equation is time dependant, the moment equations are not, as the time dependent terms in (25) and (26) are neglected because the Reynolds number is small. A time dependent simulation is then ran until the system reaches equilibrium. This equilibrium state will give us an idea of the operating temperature near the device. We impose a constant temperature of 1 at the heat source, and measure the temperature of the liquid at the heat source once equilibrium is reached, which we shall denote ω_e . All simulations are performed using the R13 equations for the vapour section, and are done at $\text{Kn} = 0.05$.

First we look at the effect of porosity on cooling. We shall fix $R_\phi = 0.6$, and show the results for three selected values of u_∞ , namely 0, 0.1 and 0.2. The results are shown in figure 19. As expected, for $u_\infty = 0$, since there is no evaporation or temperature jump, there is no heat loss in the system, so the equilibrium temperature is the temperature of the heat source. We see in figure 19a that we have an increase in ω_e with increasing ϕ , for the remaining values of u_∞ . This seems somewhat counter-intuitive at first, as we expect an increase in evaporative area to give an decrease in ω_e . However, if the free-stream velocity is fixed, then velocity of evaporative gas increases for smaller porosity. This increases the outgoing heat flux due to evaporation. The key figure we are interested in is the energy dissipated from the evaporation, which can be calculated using

$$\frac{\partial E_d}{\partial t} = - \int_{\partial C_\phi} q_n dA, \quad (53)$$

where dA denotes the surface element⁴. Integration over all time yields the total energy dissipated, and this is shown in figure 19b. This shows a decrease in dissipated energy as ϕ increases, resulting in an increase in ω_e . Furthermore, an increase in evaporation velocity yields larger temperature jumps at the interface, which, from (51), results in greater heat loss.

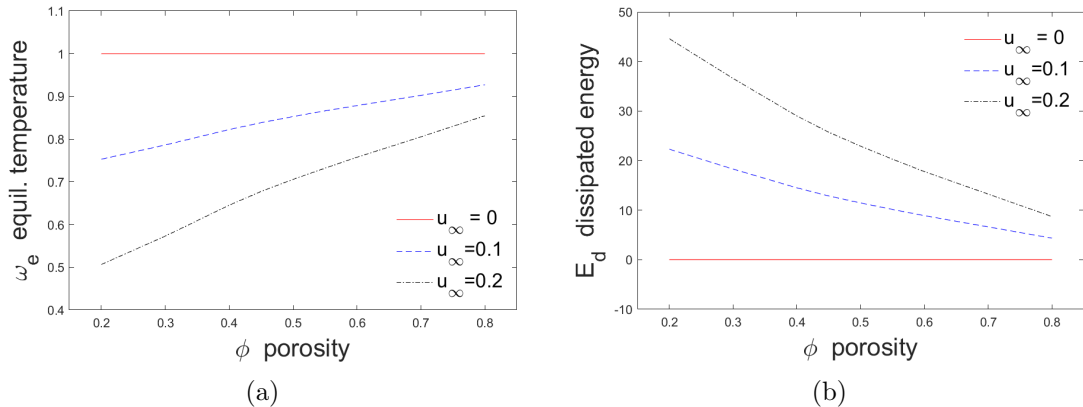


Figure 19: Effect on ω_e (left) and on E_d (right) of varying ϕ , for 3 values of u_∞ . Time ran from 0 to 1500 dimensionless time steps.

⁴our surface is a line in 2D, thus this integral is a line integral.

We observe a similar phenomenon when we look at the effect of changing the radius of the meniscus, see figure 20. An increase in radius corresponds to a flatter meniscus, thus lower surface area. So we have that decreasing the surface area actually corresponds to an increase in energy dissipated, seen in figure 20b, which in turn leads to a decrease in ω_e , seen in figure 20a. Therefore, when varying both parameters ϕ and R_ϕ individually, we observed that an increase in meniscus surface area resulted in an increase equilibrium temperature.

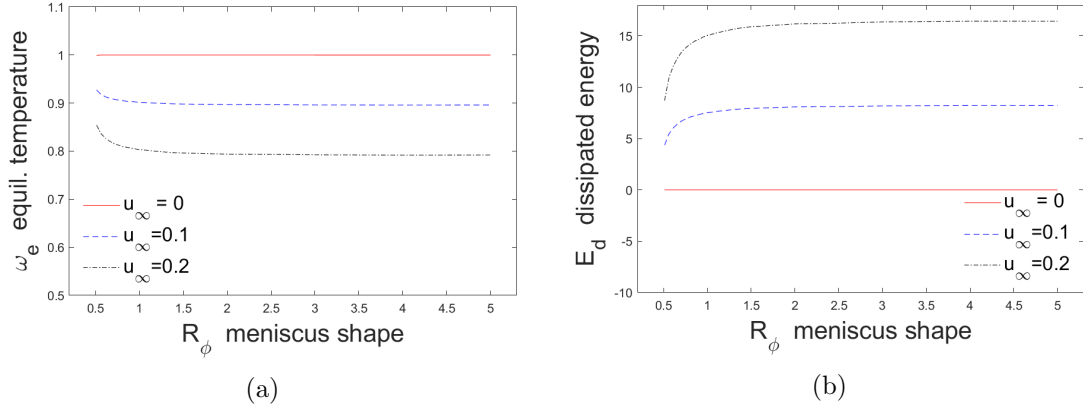


Figure 20: Effect on ω_e (left) and on E_d (right) of varying R_ϕ , for 3 values of u_∞ . Time ran from 0 to 1500 dimensionless time steps.

We now look at the effect of changing both parameters simultaneously. The surface plot for ω_e against both ϕ and R_ϕ is shown in figure 21, for fixed $u_\infty = 0.1$. Also included are the contour lines for the surface area of the meniscus⁵, i.e. the lines represent meniscus shapes with the same area. The contours of surface area do not match exactly with the contours of temperature, as there are other mechanisms at play other than just surface area, e.g. temperature jumps at the interface contribute to dissipating heat, and the heat flux across the meniscus will not be constant. However, there is a lot of resemblance between the surface area contours and the temperature contours, so that the meniscus surface area gives us a good idea of the equilibrium temperature reached. We can even use basic geometry to optimise cooling for this set up (i.e. the set up where u_∞ is fixed): if R_ϕ is large, the most effective⁶ way to increase cooling is to decrease the porosity; whereas for $R_\phi \approx 0.5$ we should increase the radius.

⁵the surface area is given by $A = R_\phi \phi \arcsin\left(\frac{1}{2R_\phi}\right)$

⁶that is, moving in the direction of steepest decent

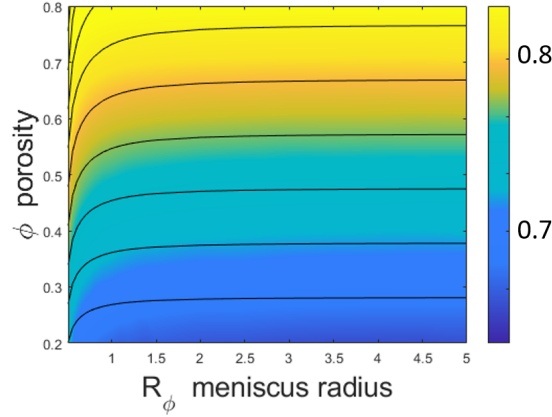


Figure 21: Surface plot for the equilibrium temperature ω , varying with ϕ and R_ϕ , $u_\infty = 0.1$. Overlaid are the contours for the menisci with constant surface area.

6 Discussion

In this discussion, we will give possible reasons as to why the data from the moment equations does not seem to match perfectly with the DSMC results in section 4. This includes considerations for the free-stream Mach number Ma_∞ , the expressions with which J was normalised with respect to, and conjectures about the parameter value jumps going from the interface into the gas. We will also show that for very small Knudsen numbers, the solutions for the moment equations are very similar, and approach the NSF solution, as expected. We then also review section 5, and give comments on how the model could be improved to give accurate data. We will also discuss the time taken for these simulations, as time reduction is one of the main reasons for studying the moment equations.

Recall that the Mach number is given as $Ma_\infty = u_\infty / \sqrt{\frac{5}{3}\theta_\infty}$, with the temperature itself dependent on the far-field velocity. The far-field velocity is the only parameter which is prescribed, and so the Mach number itself depends on the computed solution for that particular velocity. This may lead to slight inconsistencies when making comparisons between models, as an incorrect far-field temperature may exaggerate errors in other parameters.

The free-stream Mach number J is normalised with respect to the solution of the Grad13 equations, J_1 , given by (48). The DMSC data, however, has been normalised with respect to a different analytic expression, [6][7]. For the planar case the DSMC results match very well with this analytic expression, with its profile almost identical to the data in figure 10. As can be seen by the plot, for large Mach numbers, our analytic expression, J_1 , and theirs differ greatly. However, we are only considering small Mach numbers, where the two analytic expressions agree well, and so comparison between DSMC data and Comsol data is not unreasonable.

Next, recall that even for small Mach numbers, some far-field parameters do not match the DSMC results exactly. Let us conjecture as to why these discrepancies may be occurring. We study figure 17 once again. Looking only at the shape of the profiles, for larger Knudsen numbers, more moments tend to be more accurate to DSMC results. The shapes of the profiles for the R13 model look very similar to the respective DSMC results, so the moments equations seem to describe the vapour part of the geometry fairly well. Therefore, the main reason for the different far-field parameters appears to come from the jump in parameter values in the transition from liquid to vapour. For instance, since the mass flux jump value is lower for the DSMC than any of the moment equations, conservation of mass dictates that further down the line, we must have an increase in mass flux. Thus the far field mass flux must be higher to compensate. This effect is amplified as the free-stream Mach number is increased, leading to larger deviations of calculated values.

The temperature jump looks to suffer from the same issue. A close up of the temperature plot in 13 is given in figure 22, for just the flat meniscus case. The moment equations capture the linear relation between Mach number and temperature very well, especially the R13 model which has the gradient almost exact. However, the difference in temperature jump, visible in figure 17, means our far-field parameter always appear to be out by a constant.

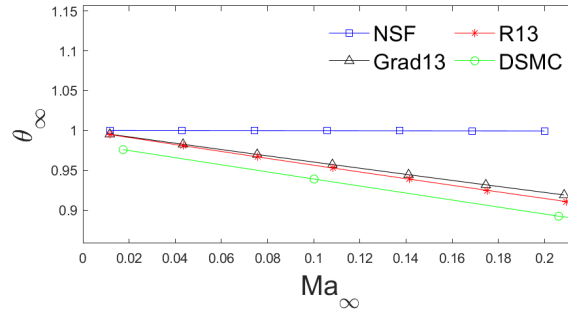


Figure 22: A closer look at far-field temperatures for all models, for the flat meniscus, $\phi = 1$, $Kn = 0.05$.

In figure 23 are the profiles for all flow parameters along the central vertical axis, for fixed $Kn=0.001$, and $u_\infty = 0.129$. This shows that NSF, Grad13 and R13 begin to converge to the same solution, which is to be expected as $Kn \rightarrow 0$. The models for the two 13 moment equations in particular show excellent agreement. The Knudsen boundary layer in this case has length 0.001, and so Knudsen boundary layer effects play a very small role.

In section 5, we gave results of a model designed to simulate the cooling of a microscale device. We did this by fixing the meniscus shape and the far-field velocity. However, the far-field velocity will depend on the meniscus size and shape. Therefore to obtain an accurate description of optimising parameters, one would need to model the pentane appropriately, using NSF say, so that the evaporation velocity is dependent on the the meniscus size and shape. This would also give accurate values for the saturation pressure

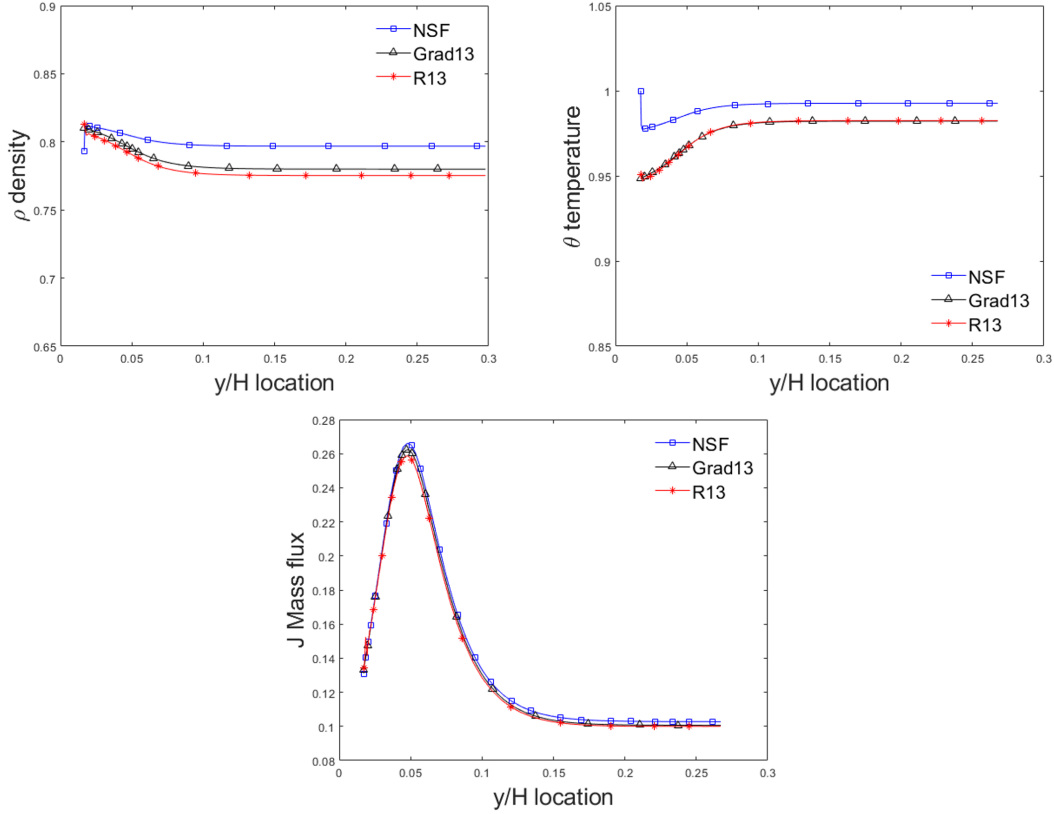


Figure 23: Flow parameters along the central vertical axis, for the semicircular meniscus case, with $\text{Kn} = 0.001$, $\phi = 0.5$ and $U_\infty = 30\text{m/s}$, corresponding to a dimensionless velocity of $u_\infty = 0.129$.

p_s at the interface, which has a key role in the evaporation velocity (even more so than the surface temperature $\theta_L = \omega_b$). Our saturation pressure was fixed at the dimensionless pressure $p_s = 1$.

We conclude this section with a short discussion on the timings of the simulations. All simulations in section ?? took no longer than 10s for both Grad13 and R13 simulations. The nano-porous membrane simulations required a much finer mesh near the interface, thus taking longer. For each of the 4 combinations of porosity and meniscus shape we gave results for, 7 free-stream Mach numbers were computed in one simulation. Computation time for each simulation took ≈ 1 minute for the Grad13 model and ≈ 5 minutes for the R13 model. This, I imagine, is much faster than the DSMC model took to run, especially for small Mach numbers. For instance, when Sharipov studied the effect of Mach number on the stochastic noise in a 2 dimensional DSMC model, small Mach number simulations took weeks to complete [10]. The implementation of the heat equation gives time dependency which takes much longer to run, moreover much time was needed as it took a while to reach equilibrium. For the porosity simulations we considered 9 values of ϕ , and 3 values of u_∞ , thus 27 configurations, taking ≈ 20 minutes. The meniscus shape simulations considered a

total of $21 \times 3 = 63$ different configurations, and taking ≈ 1 hour. Finally, the simulation comparing both the effect of ϕ and R_ϕ contained $21 \times 13 = 273$ configurations, and took ≈ 3 hours. [Include RAM data]

7 Conclusion

Models used to solve the 13 moment equations numerically were developed and tested extensively against both analytic and DSMC results. This model was designed for simulating the evaporation from a nanoporous membrane, however could potentially be adapted to accommodate different geometries without too much difficulty.

The models showed excellent agreement with analytic solutions for all Knudsen numbers and Mach numbers, even for one dimensional flows modelled in two dimensions, thus validating them for use in geometries in which analytic solutions are not possible. In the case of one dimensional evaporation into a half-space, we saw that our models were only accurate with a DSMC model for small Knudsen numbers and Mach numbers (< 0.05 and < 0.1 respectively). This set restrictions for which regimes the macroscopic models were valid. Our original assumptions in the derivation of the equations, however, reflect precisely these limitations: the moments we consider were only up to second order in the Knudsen number; moreover, we linearised the equations to make them manageable, therefore we could only consider small deviations from an equilibrium state i.e. small Mach numbers. While data figures did not match exactly with DSMC data, the key relationships between the free-stream Mach numbers and the different parameters were picked up on.

We then moved onto actually simulating the cooling of a device, implementing the remainder of the geometry with appropriate macroscopic equations, which is not possible with DSMC. A relationship between surface area and the final equilibrium temperature was found and presented. Realistic modelling however would require a more accurate set of equations, such as NSF.

We discussed the results and gave possible reasons for data not matching exactly with the DSMC data. We also saw that we can use the moment equations to bridge the gap between molecular dynamics and Navier-Stokes-Fourier, with the moment equations accurate with DSMC for low Mach and Knudsen numbers, and accurate with NSF for very low Knudsen numbers. Finally, we looked at the timings for all the simulations in this report, and concluded that calculations using the moment equations take orders of magnitude less time.

If I were to return to this work in the future, it would be to create an accurate computer model of the entire geometry, extending the work done in section 5, complete with a working NSF simulation for the pentane. Then, using real world data to accurately simulate heat-flux values produced by the micro-devices (rather than treating it as a constant dimensionless temperature of 1), a complete set of results could be presented, showing the

best configuration of parameters for porosity. The use of other liquids other than pentane could also be implemented, with varying values of surface tension, giving us data to optimise the meniscus shape.

Acknowledgements

Overall, I feel much has been accomplished in this project and it would not have even begun without the help of my supervisor Dr. Laura Cooper. Without her I would still have been figuring out how to open Comsol, and she somehow managed to teach me enough to implement the R13 equations without trouble ⁷.

I would also like to thank Dr. James Sprittles and Dr. Laura Cooper for the topic they gave me which has filled (many) hours of work, and in turn working the elegant moment equations, for which I should thank Struchtrup for helping me understand them.

⁷there was a lot of trouble

References

- [1] Kevin Robert Bagnall. *Device-level thermal analysis of GaN-based electronics*. PhD thesis, Massachusetts Institute of Technology, 2013.
- [2] Chin-Hsiang Cheng and Feng-Liang Liao. Dsmc analysis of rarefied gas flow over a rectangular cylinder at all knudsen numbers. *J. Fluids Eng.*, 122, 2000.
- [3] Paul Dawkins. The heat equation, 2009. URL <http://tutorial.math.lamar.edu/Classes/DE/TheHeatEquation.aspx>.
- [4] Changsong Ding, Gaurav Soni, Payam Bozorgi, Brian D Piorek, Carl D Meinhart, and Noel C MacDonald. A flat heat pipe architecture based on nanostructured titania. *Journal of Microelectromechanical Systems*, 19(4):878–884, 2010.
- [5] Daniel F. Hanks, Zhengmao Lu, Jay Sircar, Todd R. Salamon, Dion S. Antao, Kevin R. Bagnall, Banafsheh Barabadi, and Evelyn N. Wang. Nanoporous membrane device for ultra high heat flux thermal management. *Microsystems & Nanoengineering*, 4, December 2018.
- [6] Benzi John, Ryan Enright, James E. Sprittles, Livio Gibelli, David R. Emerson, and Duncan A. Lockerby. Numerical investigation of nanoporous evaporation using direct simulation Monte Carlo. *Phys. Rev. Fluids*, 4, November 2019.
- [7] DA Labuntsov and AP Kryukov. Analysis of intensive evaporation and condensation. *International Journal of Heat and Mass Transfer*, 22, 1979.
- [8] Zhengmao Lu, Todd R Salamon, Shankar Narayanan, Kevin R Bagnall, Daniel F Hanks, Dion S Antao, Banafsheh Barabadi, Jay Sircar, Maria E Simon, and Evelyn N Wang. Design and modeling of membrane-based evaporative cooling devices for thermal management of high heat fluxes. *IEEE transactions on components, packaging and manufacturing technology*, 6, 2016.
- [9] Ravi S Prasher, Je-Young Chang, Ioan Sauciuc, Sridhar Narasimhan, David Chau, Greg Chrysler, Alan Myers, Suzana Prstic, and Chuan Hu. Nano and micro technology-based next-generation package-level cooling solutions. *Intel Technology Journal*, 9, 2005.
- [10] Felix Sharipov. Modeling and calculations of rarefied gas flows, dsmc vs kinetic equation, 2007. URL <https://lss.fnal.gov/conf/C070709.13/Sharipov.pdf>.
- [11] Henning Struchtrup. Macroscopic transport equations for rarefied gas flows. In *Macroscopic Transport Equations for Rarefied Gas Flows*. Springer, 2005.
- [12] Henning Struchtrup and Manuel Torrilhon. Regularization of grad’s 13 moment equations: derivation and linear analysis. *Physics of Fluids*, 15(9), 2003.

- [13] Henning Struchtrup and Wolf Weiss. Temperature jump and velocity slip in the moment method. *Continuum Mechanics and Thermodynamics*, 12, April 2000.
- [14] Henning Struchtrup, Alexander Beckmann, Anirudh Rana, and Aldo Frezzotti. Evaporation boundary conditions for the R13 equations of rarefied gas dynamics. *Physics of Fluids*, September 2017.
- [15] Conan Zhang and Carlos H Hidrovo. Nanoscale wicking structures. In *ASME 2009 Heat Transfer Summer Conference collocated with the InterPACK09 and 3rd Energy Sustainability Conferences*, pages 941–954. American Society of Mechanical Engineers Digital Collection, 2009.